

Principles of Anatomy and Physiology

Gerard Tortora and Bryan Derrickson

Sixteenth Edition

Chapter 4

The Tissue Level of Organization

Introduction

- The purpose of this chapter is to:
 - Describe the various types of tissues and their origins
 - Discuss how cells of a tissue are held together
 - Compare and contrast epithelial, connective, muscular, and nervous tissue
 - Learn about the structure and function of membranes
 - Understand tissue repair

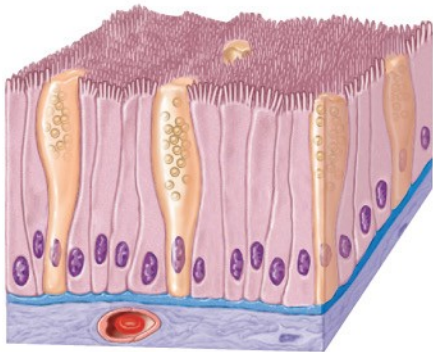
Types of Tissues

What Is a Tissue?

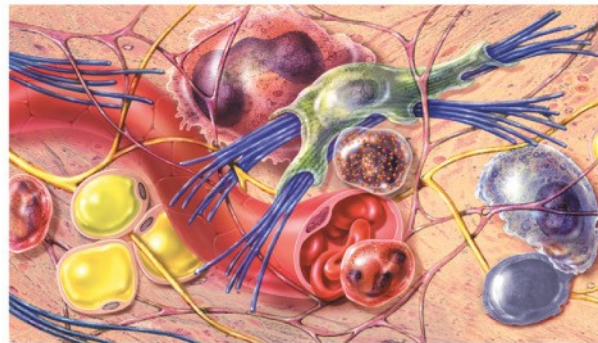
- A tissue is a group of cells that usually have a common embryonic origin and function together to carry out specialized activities
- There are 4 basic types of tissues in the human body and they are categorized according to their structure and function

The 4 Tissue Types

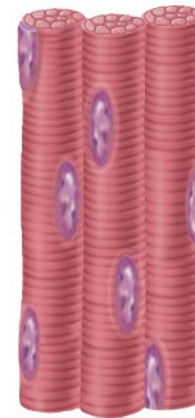
Epithelial, Connective, Muscular, Nervous



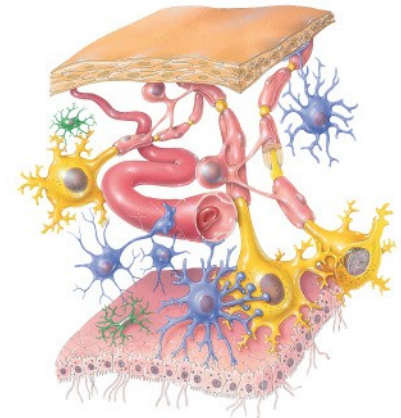
(a) Epithelial tissue



(b) Connective tissue



(c) Muscular tissue

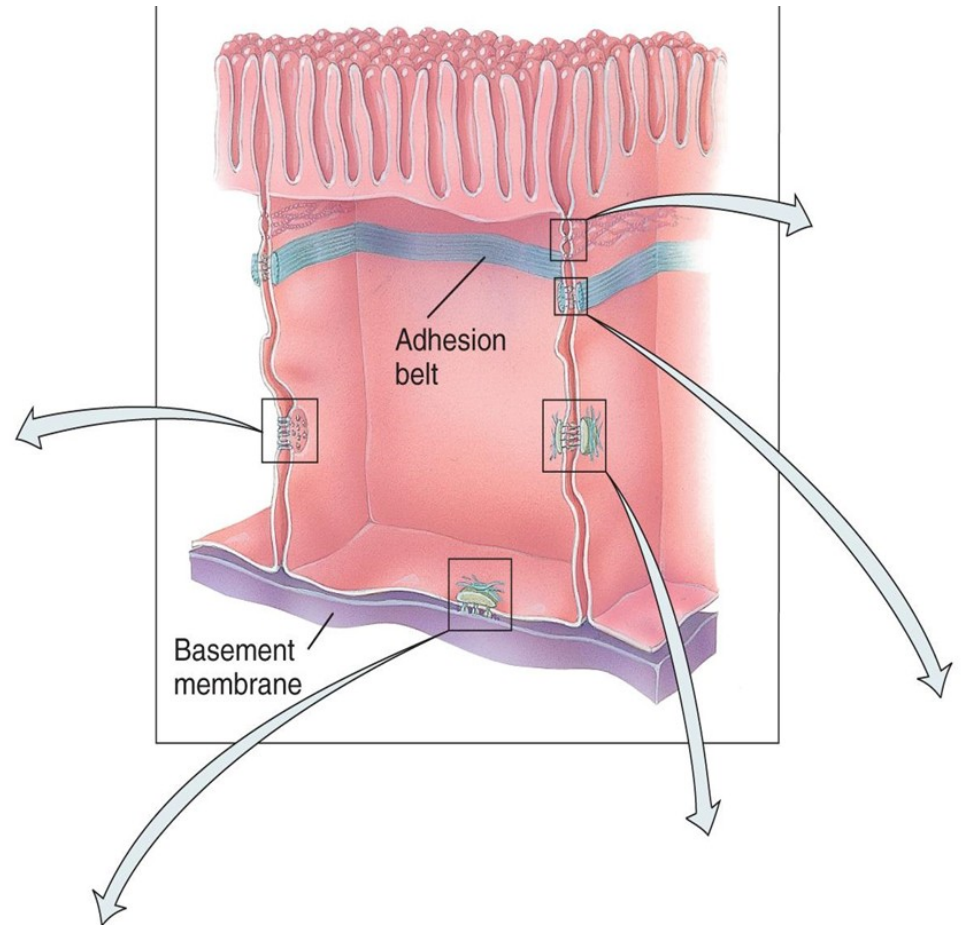


(d) Nervous tissue

Cell Junctions

Cell Junctions

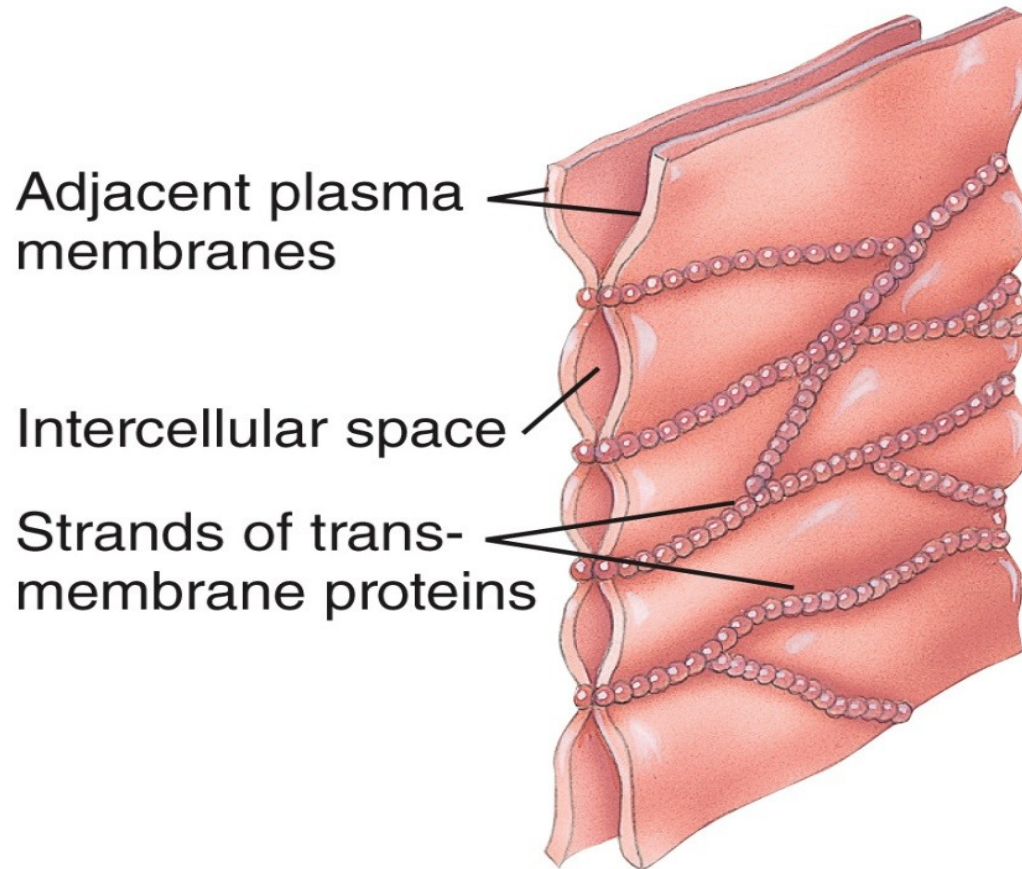
- Cells can be held together in a number of ways
- These points of contact between cells are called cell functions



Intercellular Junction Interactions Animation:

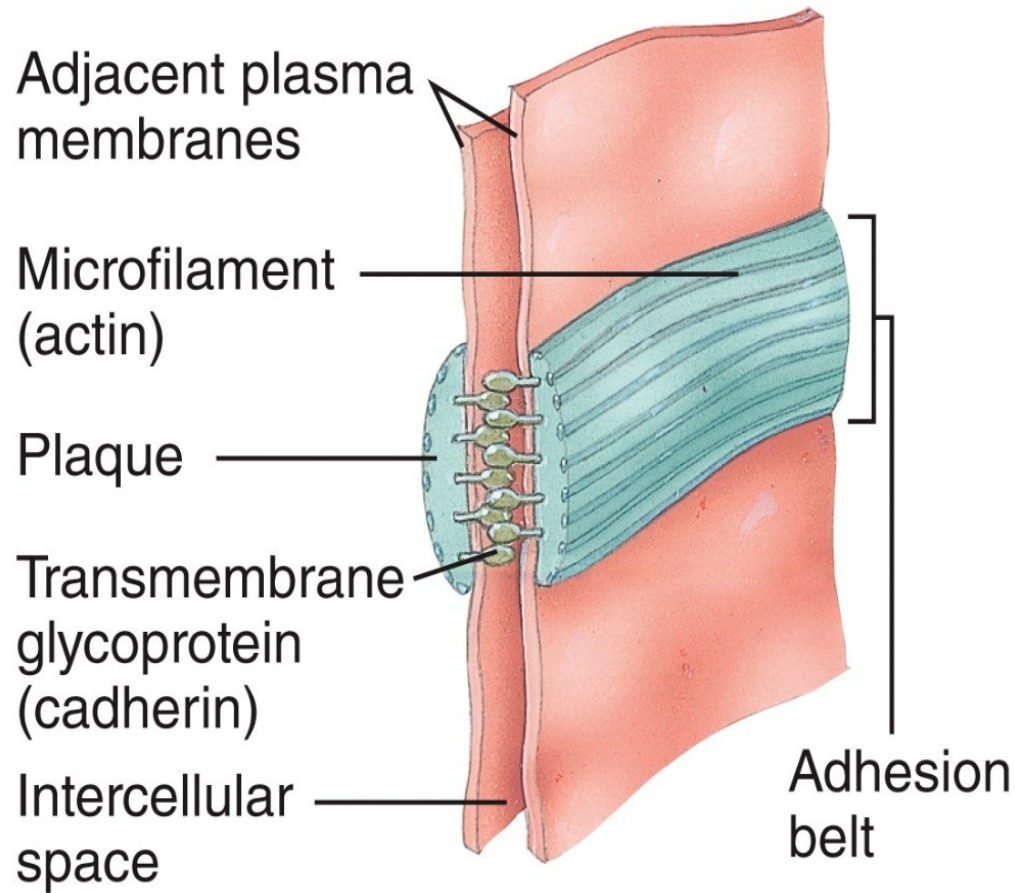
Intercellular Junctions

Tight Junctions



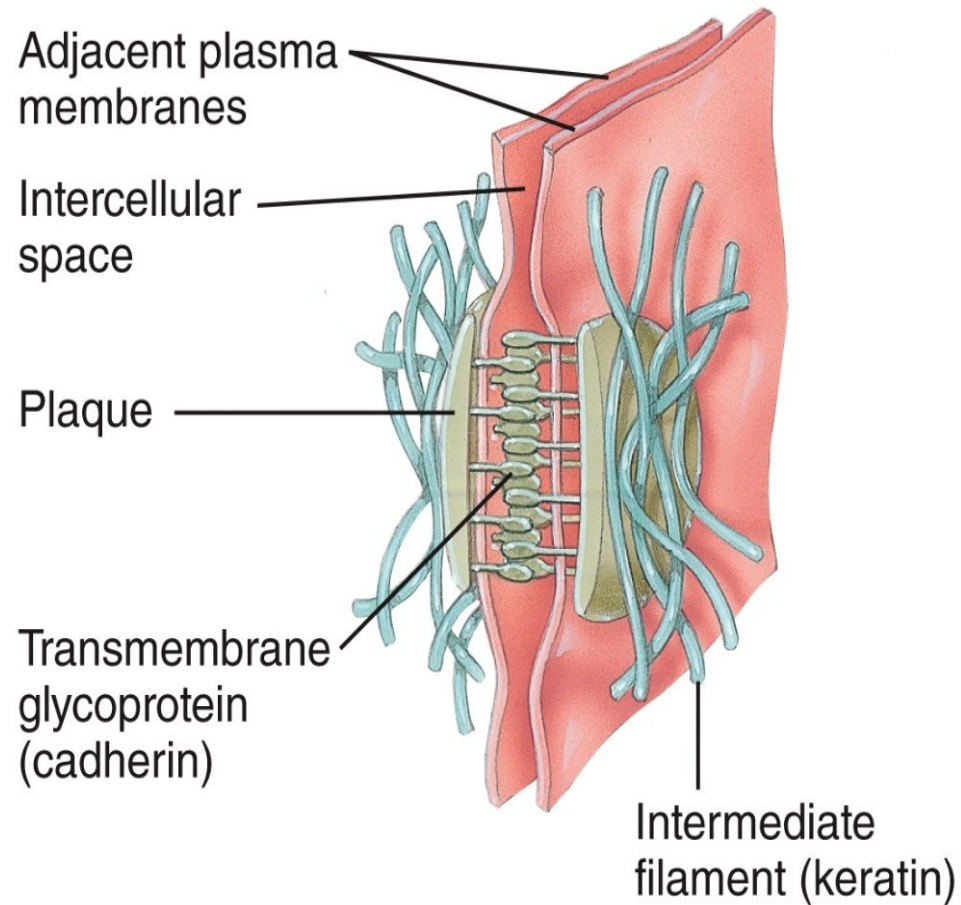
Tight Junctions

Adhering Junctions



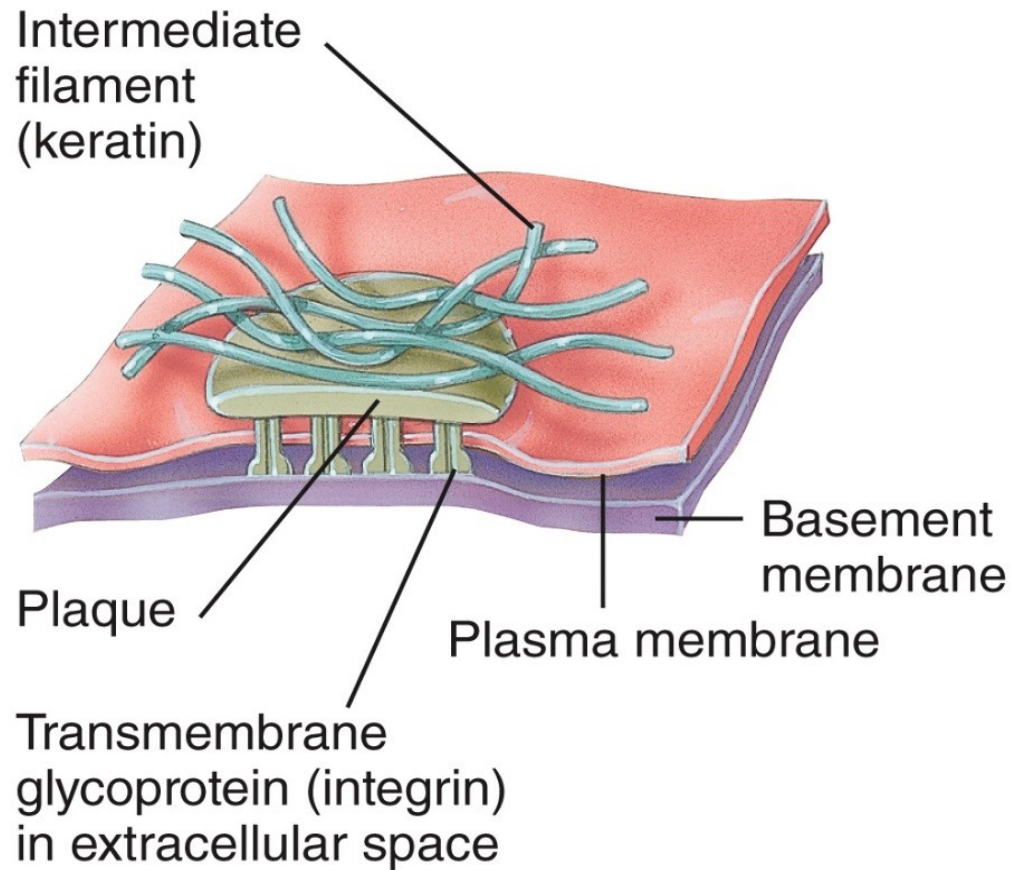
Adhering junction

Desmosomes



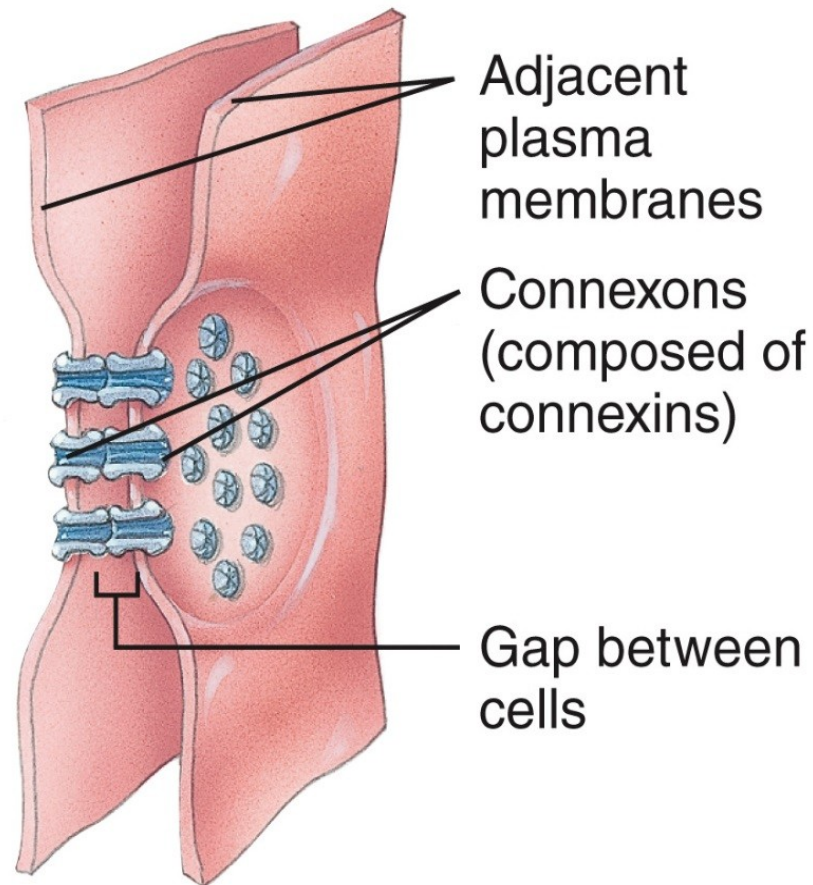
Desmosome

Hemidesmosomes



Hemidesmosome

Gap Junctions



Gap junction

Epithelial Tissue

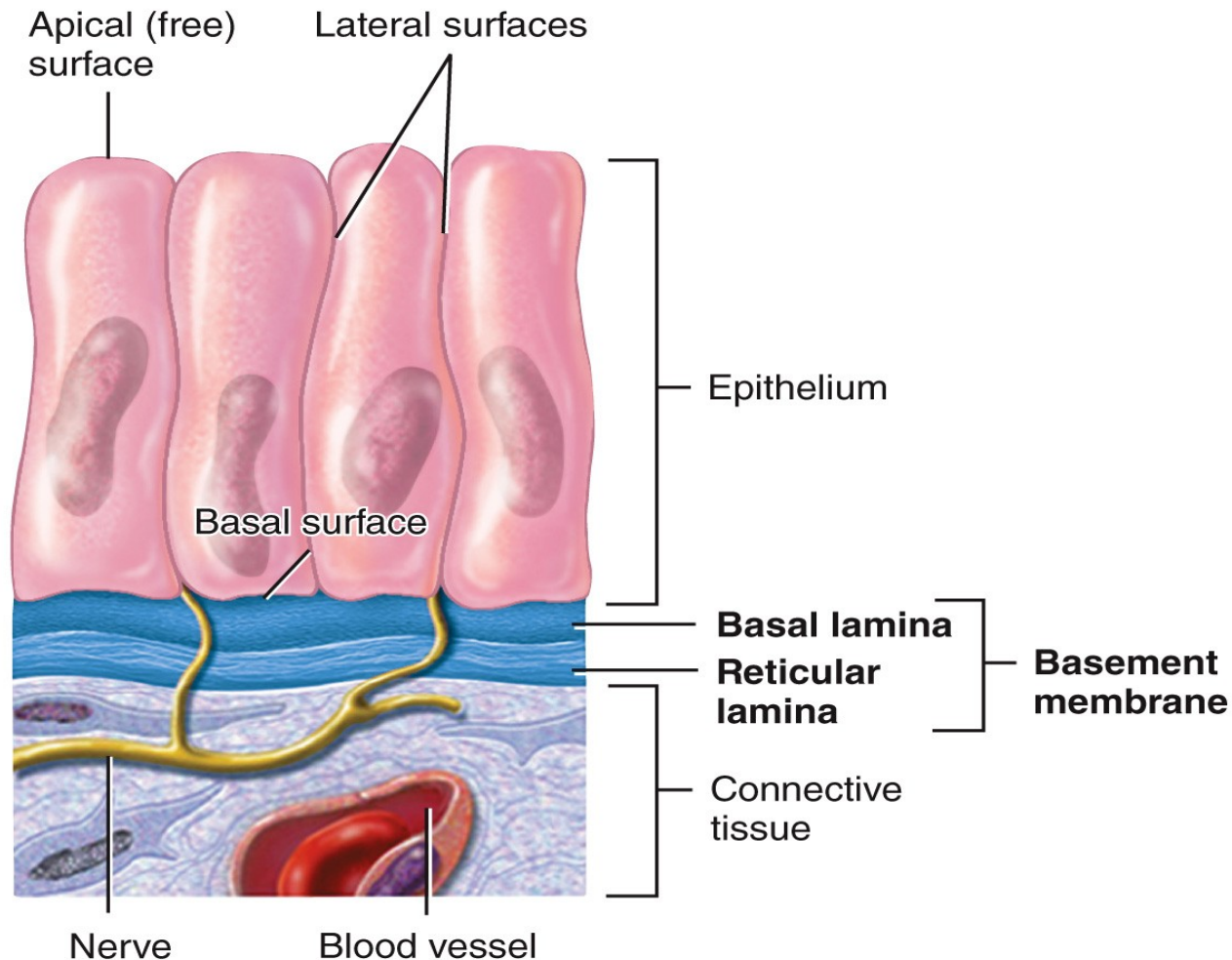
General Features of Epithelial Tissue

- Cells are arranged in sheets and are densely packed
- Many cell junctions are present
- Cells attach to a basement membrane
- Avascular but does have a nerve supply
- Mitosis occurs frequently

General Features of Epithelial Tissue

- Tissue has polarity – apical (upper, free) and basal (lower, attached) surfaces
 - Apical surfaces may bear microvilli (ex: brush border of intestinal lining) which increases surface area of plasma membrane for absorption or cilia (ex: lining trachea) to sweep substances across its surface
- Two main types, by location:
 - Covering and lining
 - On external surfaces or lining cavities
 - Glandular
 - Secretory tissue in glands

Surfaces of Epithelial Cells and the Basement Membrane

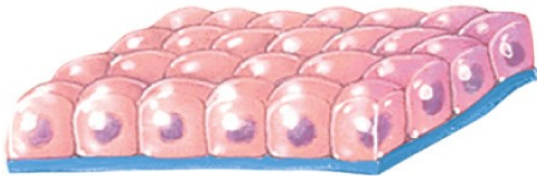


Classification of Epithelial Tissue

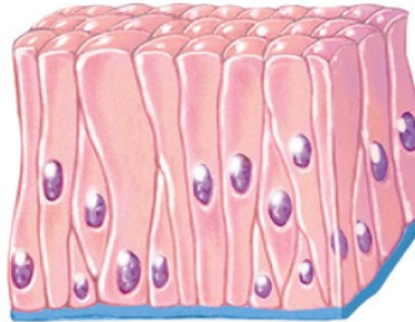
- Based on two criteria:
 - Number of layers
 - Simple – one layer
 - Stratified – multiple layers
 - Pseudostratified – one layer, but appears to be multiple
 - Cell shape
 - Squamous – flattened cells
 - Cuboidal – cube-shaped cells
 - Columnar – upright rectangular
 - *when stratified, cell shape is based on layer at apical surface

Classification of Epithelial Tissue

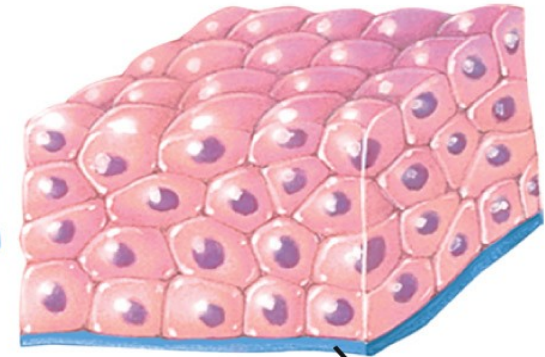
Arrangement of layers



Simple



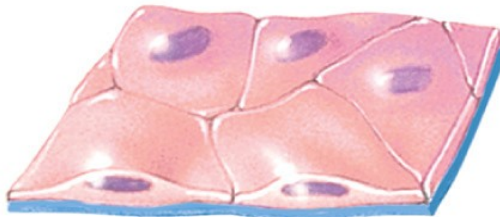
Pseudostratified



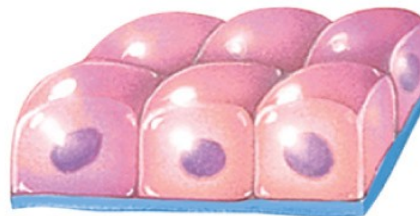
Stratified

Basement
membrane

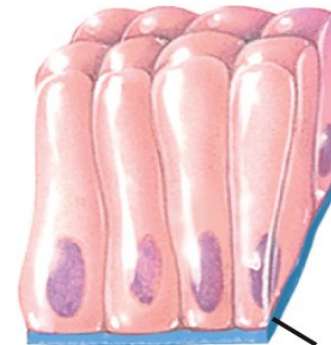
Cell shape



Squamous



Cuboidal



Columnar

Basement
membrane

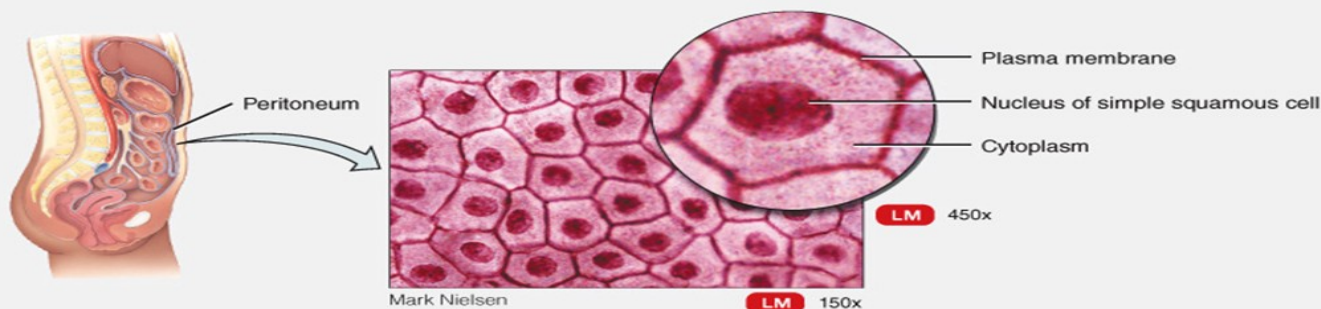
Epithelial Tissue Anatomy Overview:

Epithelial Tissues

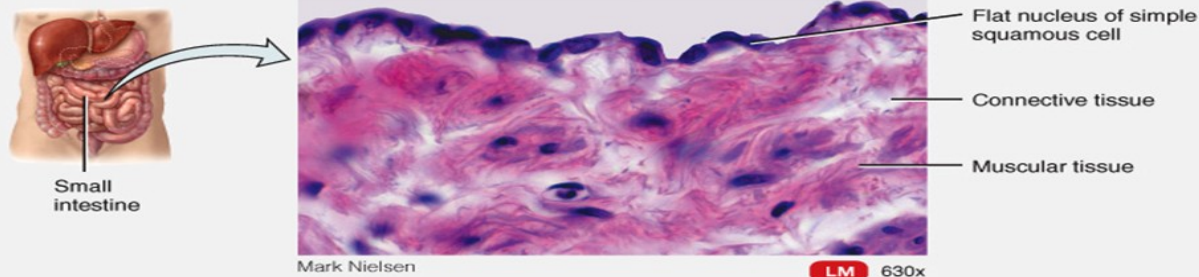
TABLE 4.1 Epithelial Tissue: Covering and Lining Epithelium

A. SIMPLE SQUAMOUS EPITHELIUM

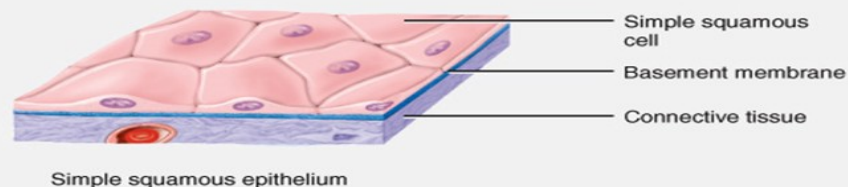
Description	Simple squamous epithelium is a single layer of flat cells that resembles a tiled floor when viewed from apical surface; centrally located nucleus that is flattened and oval or spherical in shape.
Location	Most commonly (1) lines the cardiovascular and lymphatic system (heart, blood vessels, lymphatic vessels), where it is known as endothelium (en'-dō-THĒ-lē-um; <i>endo-</i> = within; <i>-thelium</i> = covering), and (2) forms the epithelial layer of serous membranes (peritoneum, pleura, pericardium), where it is called mesothelium (mez'-ō-THĒ-lē-um; <i>meso-</i> = middle). Also found in air sacs of lungs, glomerular (Bowman's) capsule of kidneys, inner surface of tympanic membrane (eardrum).
Function	Present at sites of filtration (such as blood filtration in kidneys) or diffusion (such as diffusion of oxygen into blood vessels of lungs) and at site of secretion in serous membranes. Not found in body areas subject to mechanical stress (wear and tear).



Surface view of simple squamous epithelium of mesothelial lining of peritoneum



Sectional view of simple squamous epithelium (mesothelium) of peritoneum of small intestine



Simple squamous epithelium

B. SIMPLE CUBOIDAL EPITHELIUM

Description

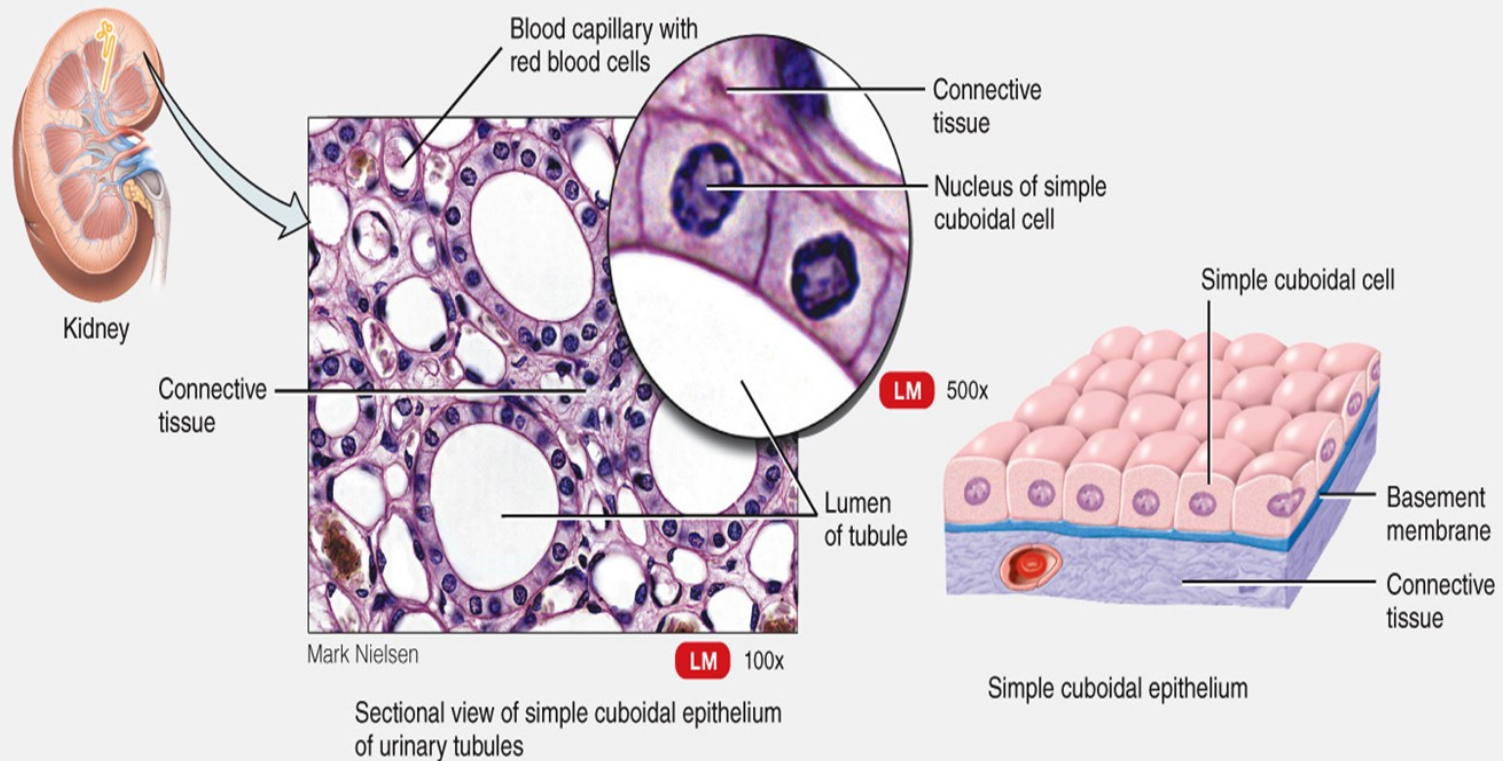
Simple cuboidal epithelium is a single layer of cube-shaped cells; round, centrally located nucleus. Cuboidal cell shape is obvious when tissue is sectioned and viewed from the side. (Note: Strictly cuboidal cells could not form small tubes; these cuboidal cells are more pie-shaped but still nearly as high as they are wide at the base.)

Location

Covers surface of ovary; lines anterior surface of capsule of lens of the eye; forms pigmented epithelium at posterior surface of retina of the eye; lines kidney tubules and smaller ducts of many glands; makes up secreting portion of some glands such as thyroid gland and ducts of some glands such as pancreas.

Function

Secretion and absorption.



C. NONCILIATED SIMPLE COLUMNAR EPITHELIUM

Description **Nonciliated simple columnar epithelium** is a single layer of nonciliated columnlike cells with oval nuclei near base of cells; contains (1) columnar epithelial cells with microvilli at apical surface and (2) goblet cells. **Microvilli**, fingerlike cytoplasmic projections, increase surface area of plasma membrane (see [Figure 3.1](#)), thus increasing cell's rate of absorption. **Goblet cells** are modified columnar epithelial cells that secrete mucus, a slightly sticky fluid, at their apical surfaces. Before release, mucus accumulates in upper portion of cell, causing it to bulge and making the whole cell resemble a goblet or wine glass.

Location Lines gastrointestinal tract (from stomach to anus), ducts of many glands, and gallbladder.

Function Secretion and absorption; larger columnar cells contain more organelles and thus are capable of higher level of secretion and absorption than are cuboidal cells. Secreted mucus lubricates linings of digestive, respiratory, and reproductive tracts, and most of urinary tract; helps prevent destruction of stomach lining by acidic gastric juice secreted by stomach.

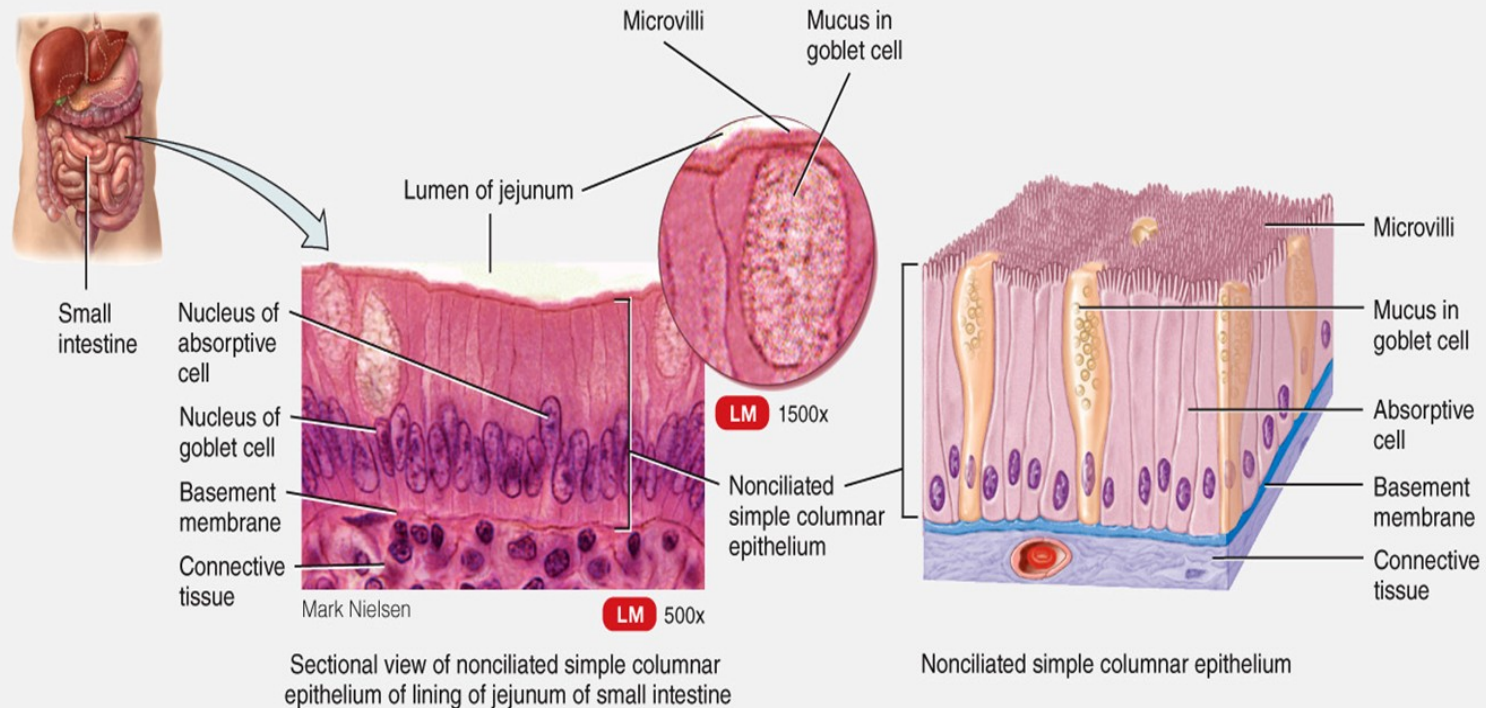
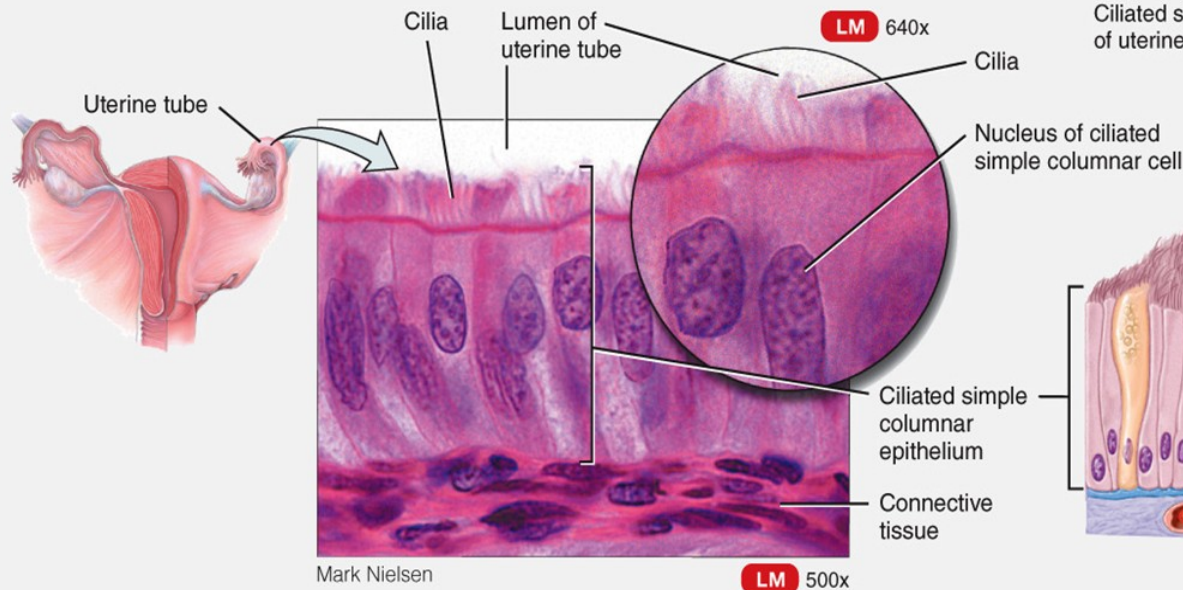
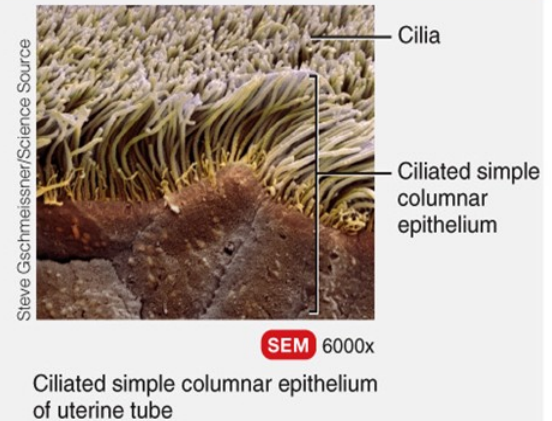


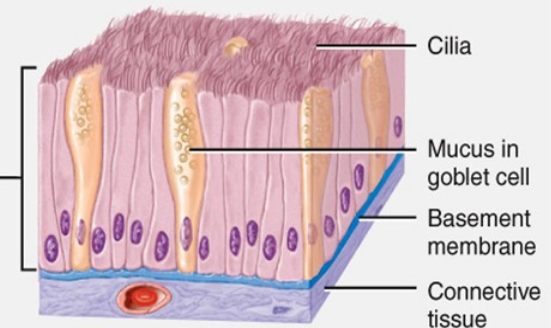
TABLE 4.1 Epithelial Tissue: Covering and Lining Epithelium (*Continued*)

D. CILIATED SIMPLE COLUMNAR EPITHELIUM

Description	Ciliated simple columnar epithelium is a single layer of ciliated columnlike cells with oval nuclei near base of cells. Goblet cells are usually interspersed.
Location	Lines some bronchioles (small tubes) of respiratory tract, uterine (fallopian) tubes, uterus, some paranasal sinuses, central canal of spinal cord, and ventricles of brain.
Function	Cilia beat in unison, moving mucus and foreign particles toward throat, where they can be coughed up and swallowed or spit out. Coughing and sneezing speed up movement of cilia and mucus. Cilia also help move oocytes expelled from ovaries through uterine (fallopian) tubes into uterus.



Sectional view of ciliated simple columnar epithelium of uterine tube

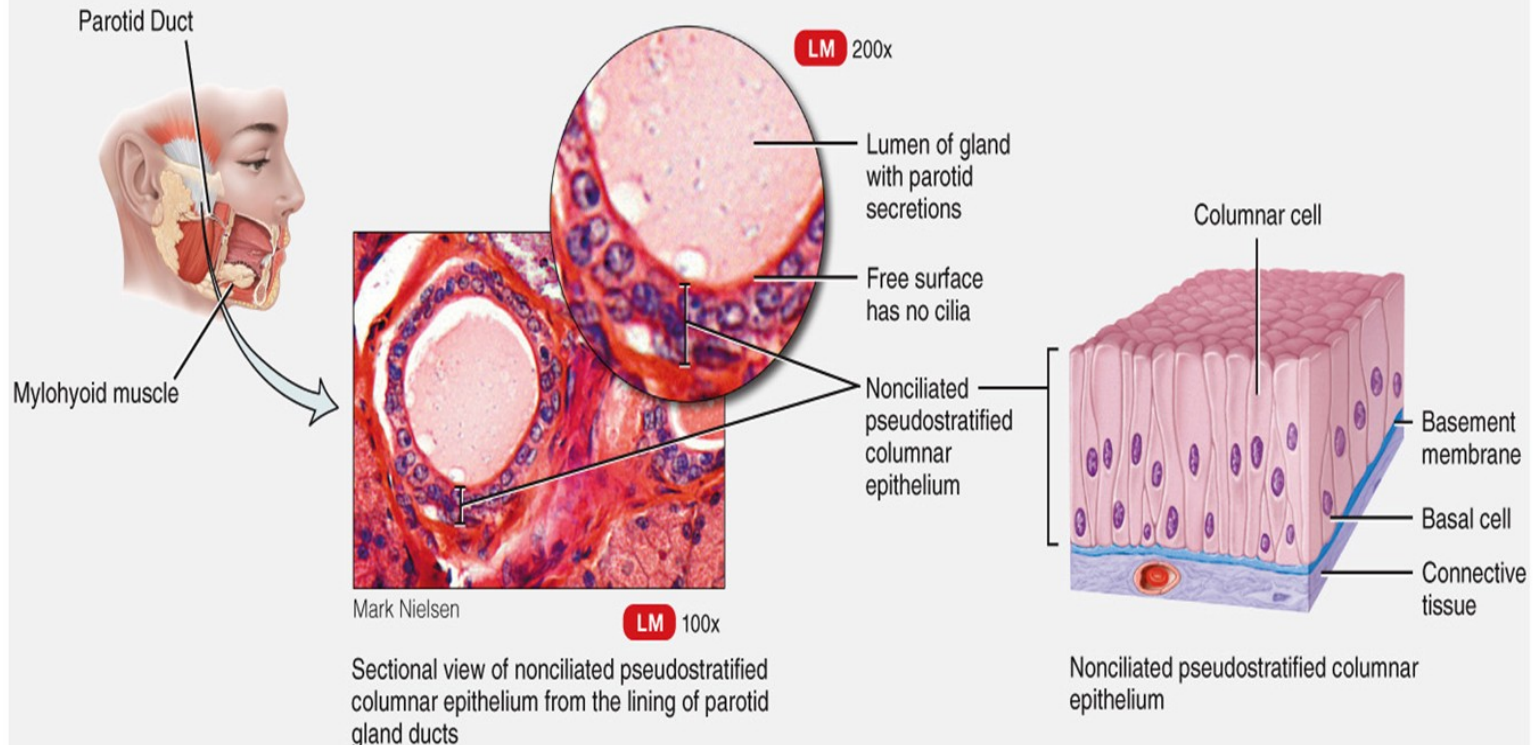


Ciliated simple columnar epithelium

TABLE 4.1 Epithelial Tissue: Covering and Lining Epithelium (*Continued*)

E. NONCILIATED PSEUDOSTRATIFIED COLUMNAR EPITHELIUM

Description	Nonciliated pseudostratified columnar epithelium appears to have several layers because the nuclei of the cells are at various levels. Even though all the cells are attached to the basement membrane in a single layer, some cells do not extend to the apical surface. When viewed from the side, these features give the false impression of a multilayered tissue—thus the name pseudostratified epithelium (<i>pseudo-</i> = false). Contains cells without cilia and also lacks goblet cells.
Location	Lines epididymis, larger ducts of many glands, and parts of male urethra.
Function	Absorption and secretion.



F. CILIATED PSEUDOSTRATIFIED COLUMNAR EPITHELIUM

Description

Ciliated pseudostratified columnar epithelium appears to have several layers because cell nuclei are at various levels. All cells are attached to basement membrane in a single layer, but some cells do not extend to apical surface. When viewed from side, these features give false impression of a multilayered tissue (thus the name pseudostratified; *pseudo* = false). Contains cells that extend to surface and secrete mucus (goblet cells) or bear cilia.

Location

Lines airways of most of upper respiratory tract.

Function

Secretes mucus that traps foreign particles, and cilia sweep away mucus for elimination from body.

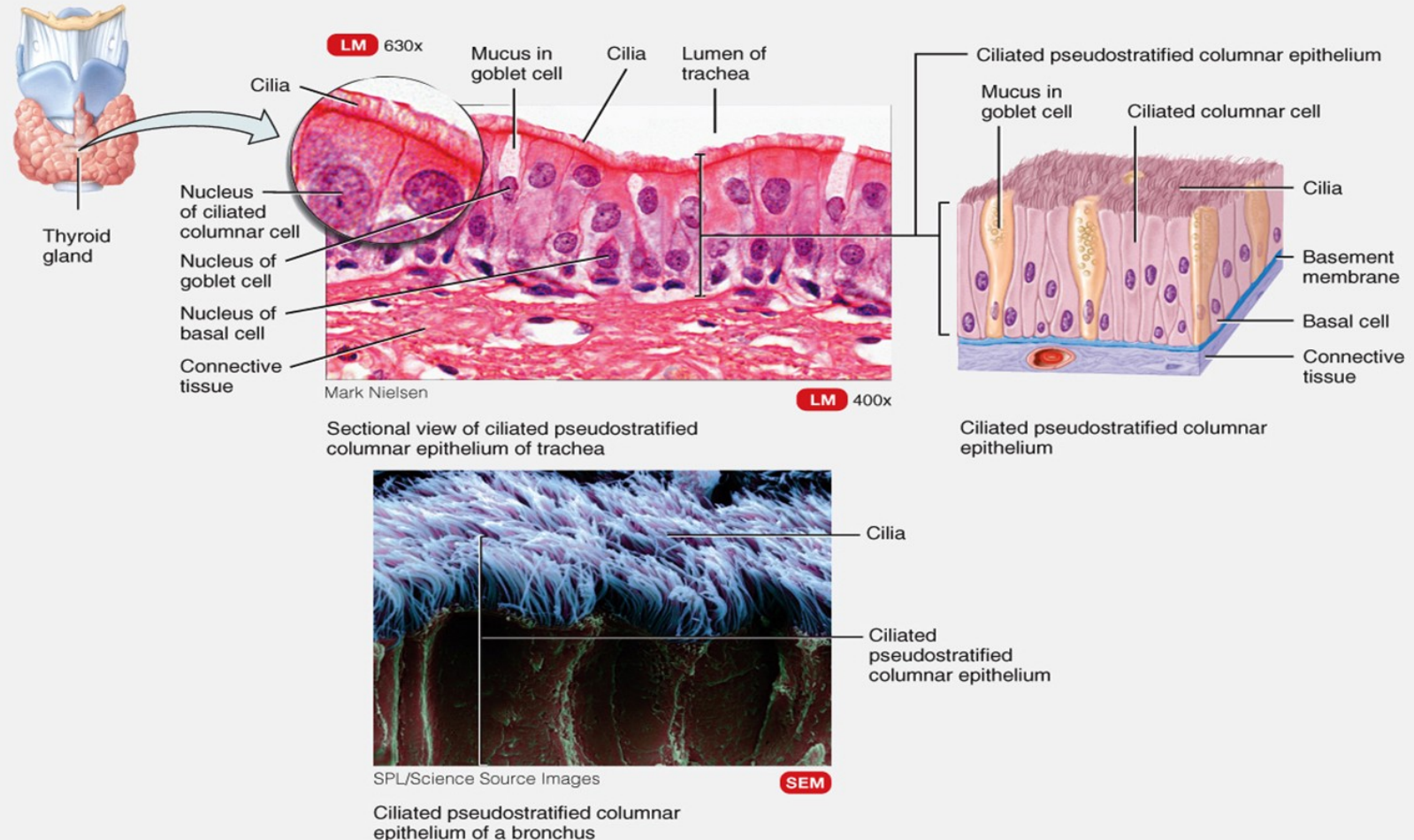


TABLE 4.1 Epithelial Tissue: Covering and Lining Epithelium (*Continued*)

G. STRATIFIED SQUAMOUS EPITHELIUM

Description

Stratified squamous epithelium has two or more layers of cells; cells in apical layer and several layers deep to it are squamous; cells in deeper layers vary from cuboidal to columnar. As basal cells divide, daughter cells arising from cell divisions push upward toward apical layer. As they move toward surface and away from blood supply in underlying connective tissue, they become dehydrated and less metabolically active. Tough proteins predominate as cytoplasm is reduced, and cells become tough, hard structures that eventually die. At apical layer, after dead cells lose cell junctions they are sloughed off, but they are replaced continuously as new cells emerge from basal cells.

Keratinized stratified squamous epithelium develops tough layer of keratin in apical layer of cells and several layers deep to it (see [Figure 5.3](#)). (**Keratin** is a tough, fibrous intracellular protein that helps protect skin and underlying tissues from heat, microbes, and chemicals.) Relative amount of keratin increases in cells as they move away from nutritive blood supply and organelles die.

Nonkeratinized stratified squamous epithelium does not contain large amounts of keratin in apical layer and several layers deep and is constantly moistened by mucus from salivary and mucous glands; organelles are not replaced.

Location

Keratinized variety forms superficial layer of skin; nonkeratinized variety lines wet surfaces (lining of mouth, esophagus, part of epiglottis, part of pharynx, and vagina) and covers tongue.

Function

Protection against abrasion, water loss, ultraviolet radiation, and foreign invasion. Both types form first line of defense against microbes.

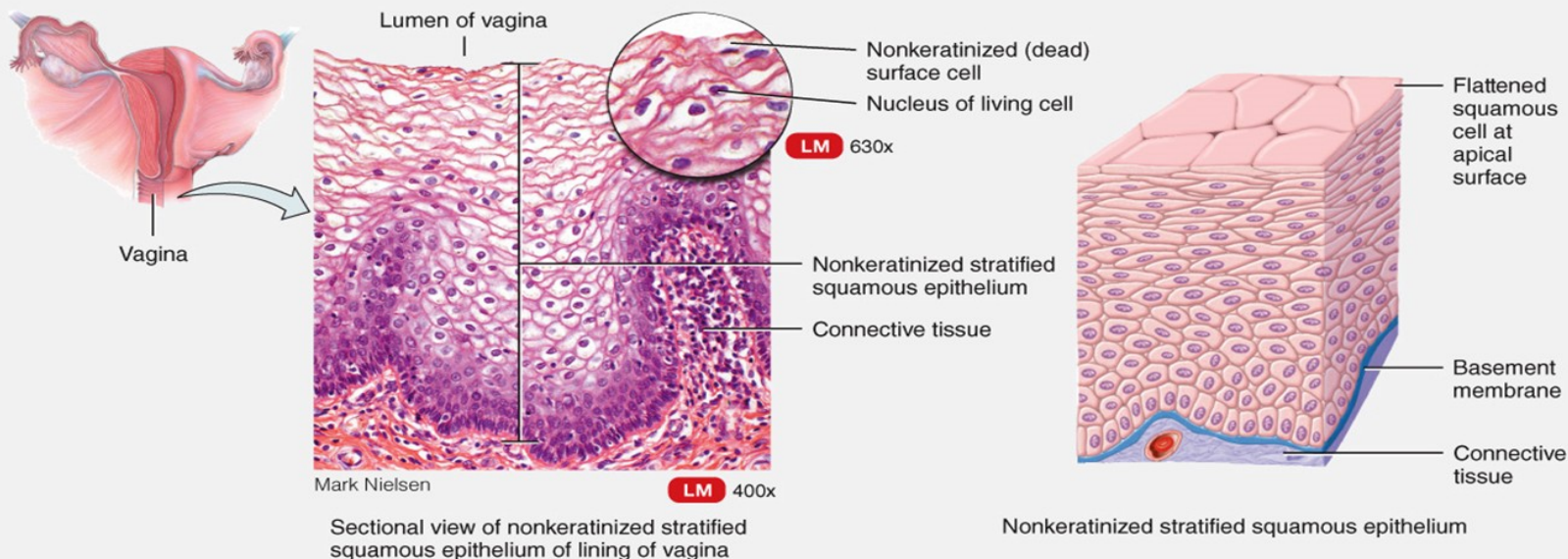


TABLE 4.1 Epithelial Tissue: Covering and Lining Epithelium (*Continued*)

G. STRATIFIED SQUAMOUS EPITHELIUM

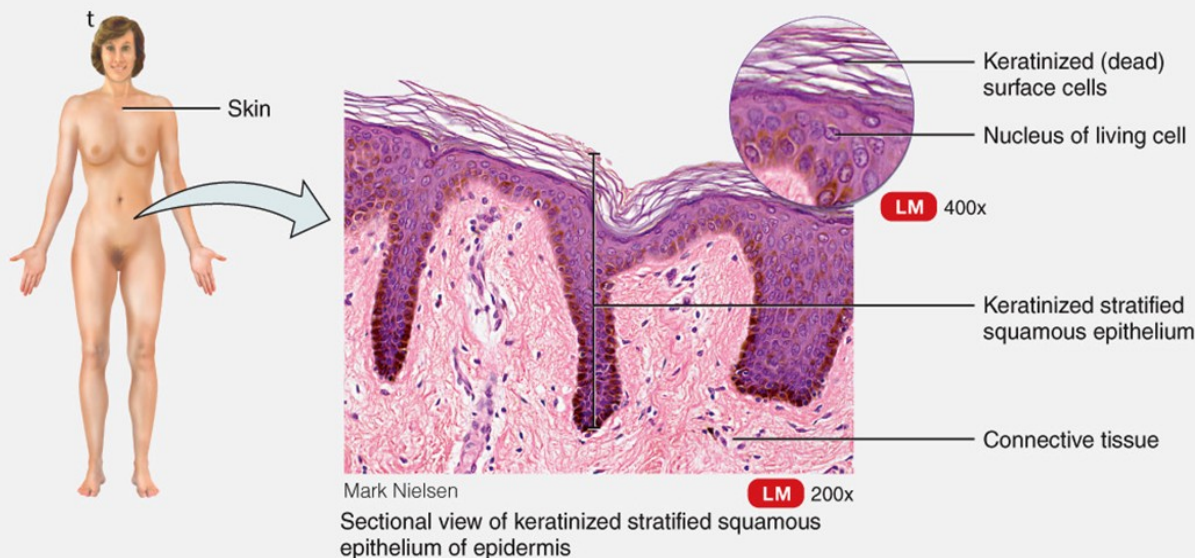
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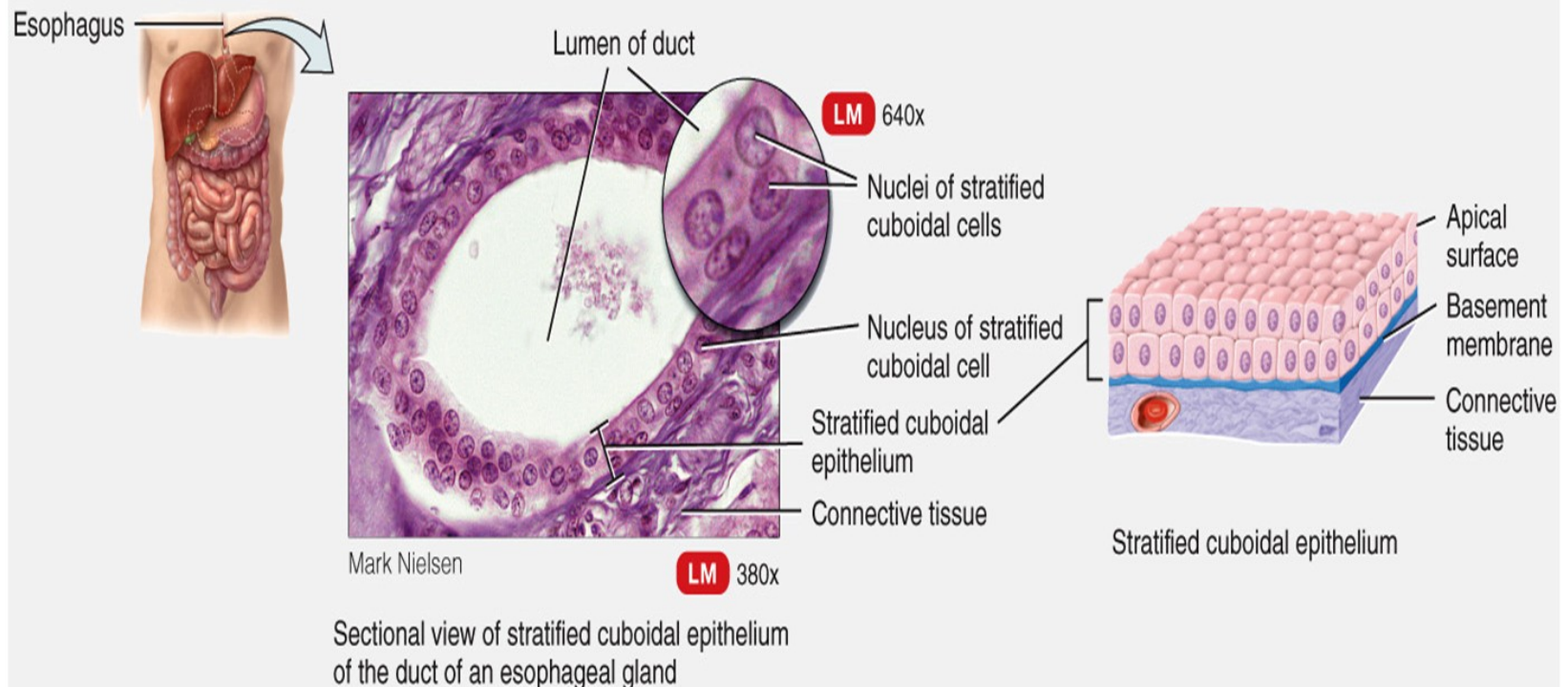
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Function Protection against abrasion, water loss, ultraviolet radiation, and foreign invasion. Both types form first line of defense against microbes.



H. STRATIFIED CUBOIDAL EPITHELIUM

Description	Stratified cuboidal epithelium has two or more layers of cells; cells in apical layer are cube-shaped; fairly rare type.
Location	Ducts of adult sweat glands and esophageal glands, part of male urethra.
Function	Protection; limited secretion and absorption.



I. STRATIFIED COLUMNAR EPITHELIUM

Description	Basal layers in stratified columnar epithelium usually consist of shortened, irregularly shaped cells; only apical layer has columnar cells; uncommon.
Location	Lines part of urethra; large excretory ducts of some glands, such as esophageal glands; small areas in anal mucous membrane; part of conjunctiva of eye.
Function	Protection and secretion.

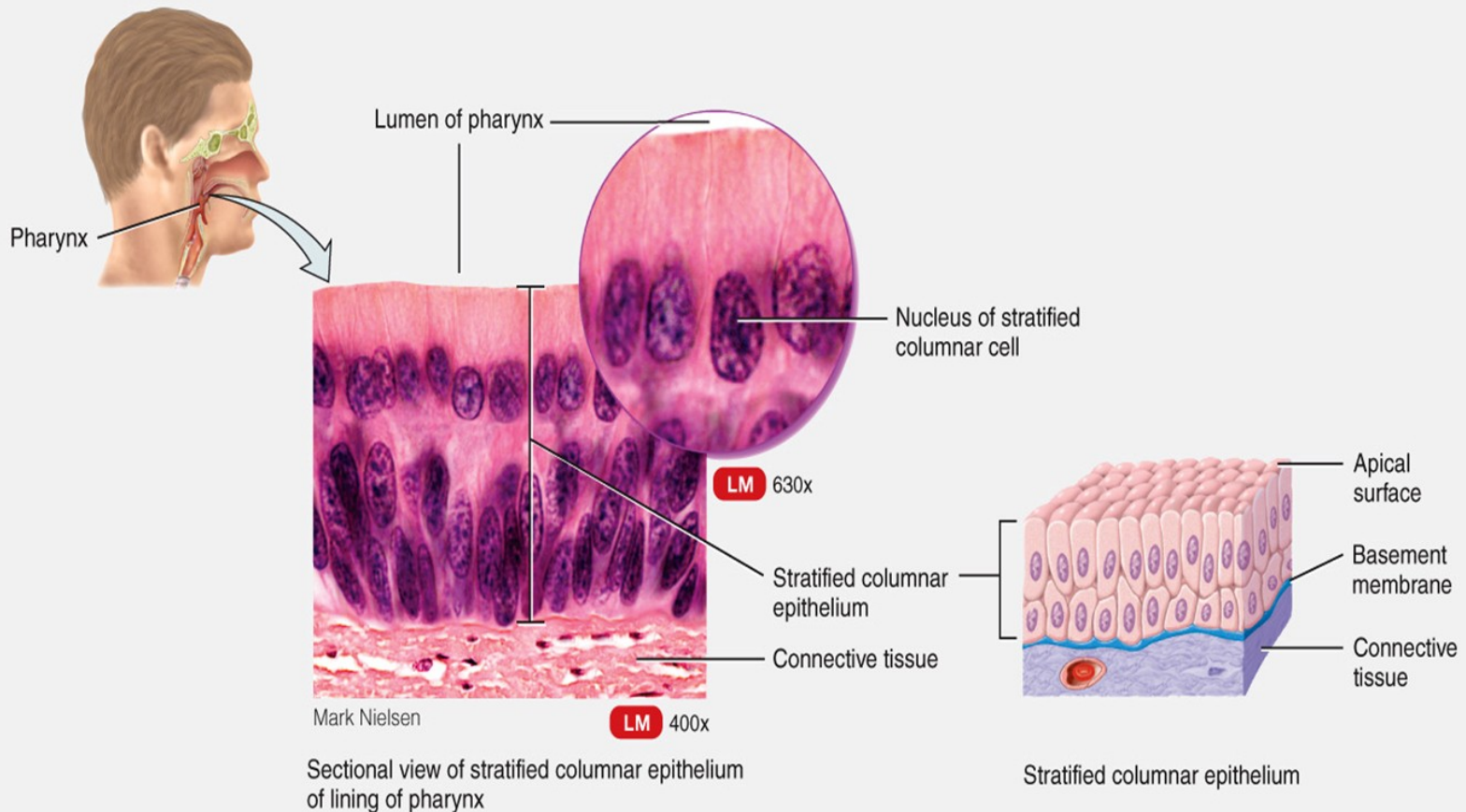


TABLE 4.1 Epithelial Tissue: Covering and Lining Epithelium (*Continued*)

J. TRANSITIONAL EPITHELIUM (UROTHELIUM)

Description **Transitional epithelium (urothelium)** has a variable appearance (transitional). In relaxed or unstretched state, looks like stratified cuboidal epithelium, except apical layer cells tend to be large and rounded. As tissue is stretched, cells become flatter, giving the appearance of stratified squamous epithelium. Multiple layers and elasticity make it ideal for lining hollow structures (urinary bladder) subject to expansion from within.

Location Lines urinary bladder and portions of ureters and urethra.

Function Allows urinary organs to stretch and maintain protective lining while holding variable amounts of fluid without rupturing.

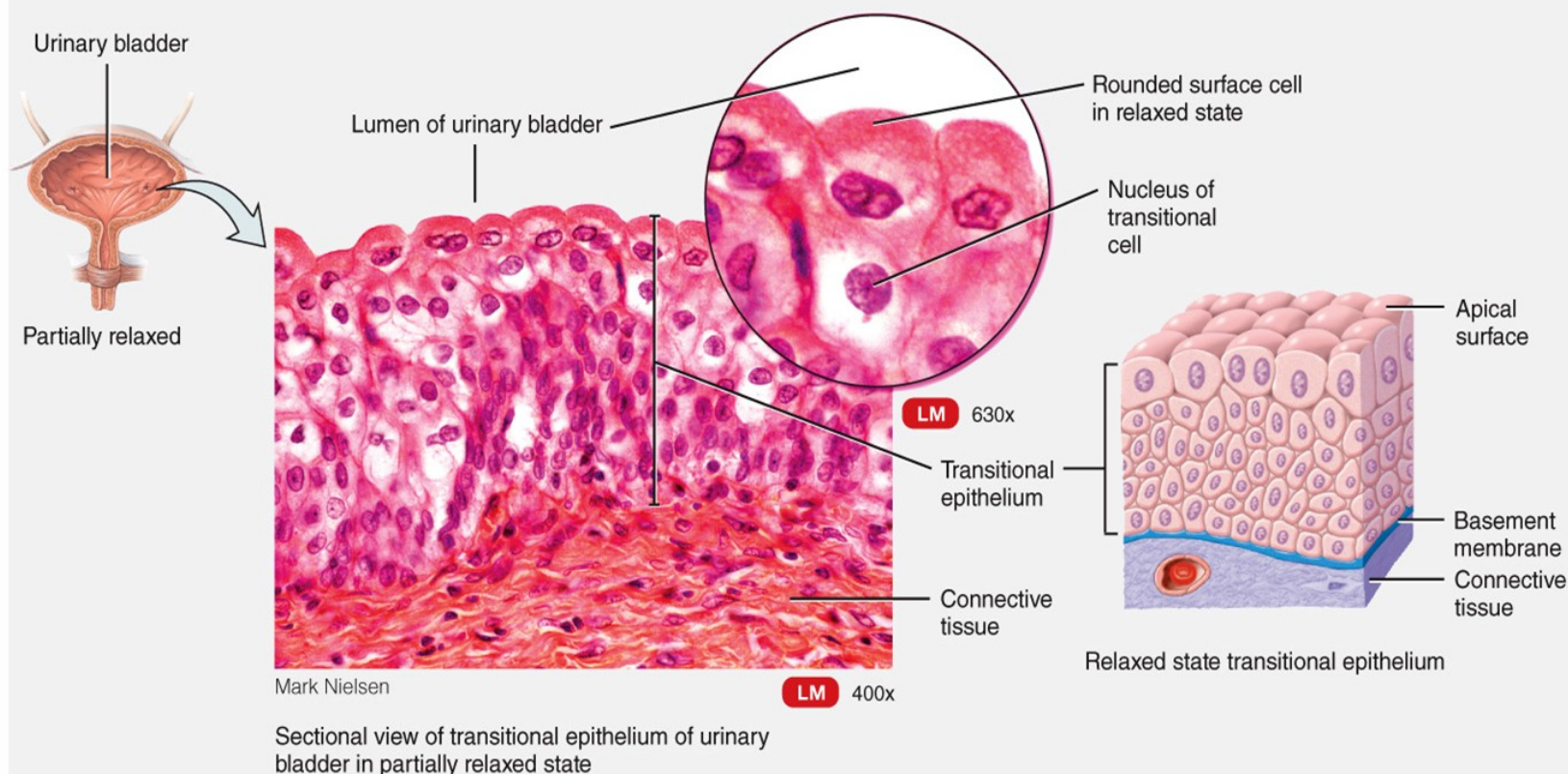


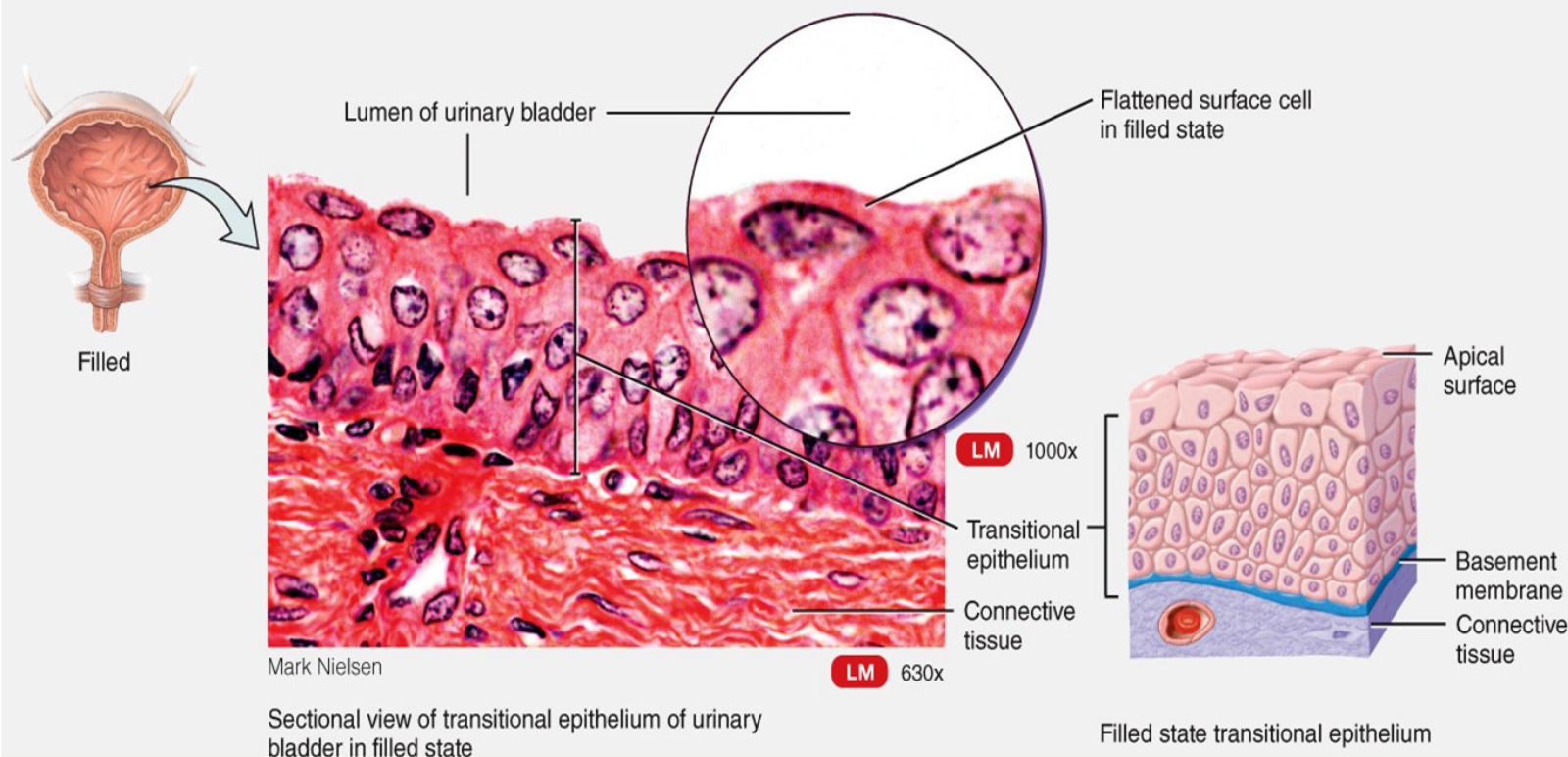
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Location Lines urinary bladder and portions of ureters and urethra.

Function Allows urinary organs to stretch and maintain protective lining while holding variable amounts of fluid without rupturing.



Glandular Epithelium

- A gland is a single cell or a mass of epithelial cells adapted for secretion
 - The secretions of endocrine glands enter the interstitial fluid and diffuse into the bloodstream
 - Endocrine glands are ductless when mature
 - The secretions of exocrine glands enter ducts that empty onto the surface of a covering/lining epithelium such as the skin surface or the lumen of the stomach

TABLE 4.2 Epithelial Tissue: Glandular Epithelium

A. ENDOCRINE GLANDS

Description	Endocrine gland secretions (<i>hormones</i>) enter interstitial fluid and then diffuse into bloodstream without flowing through a duct. Endocrine glands will be described in detail in Chapter 18.
Location	Examples include pituitary gland at base of brain, pineal gland in brain, thyroid and parathyroid glands near larynx (voice box), adrenal glands superior to kidneys, pancreas near stomach, ovaries in pelvic cavity, testes in scrotum, thymus in thoracic cavity.
Function	Hormones regulate many metabolic and physiological activities to maintain homeostasis.

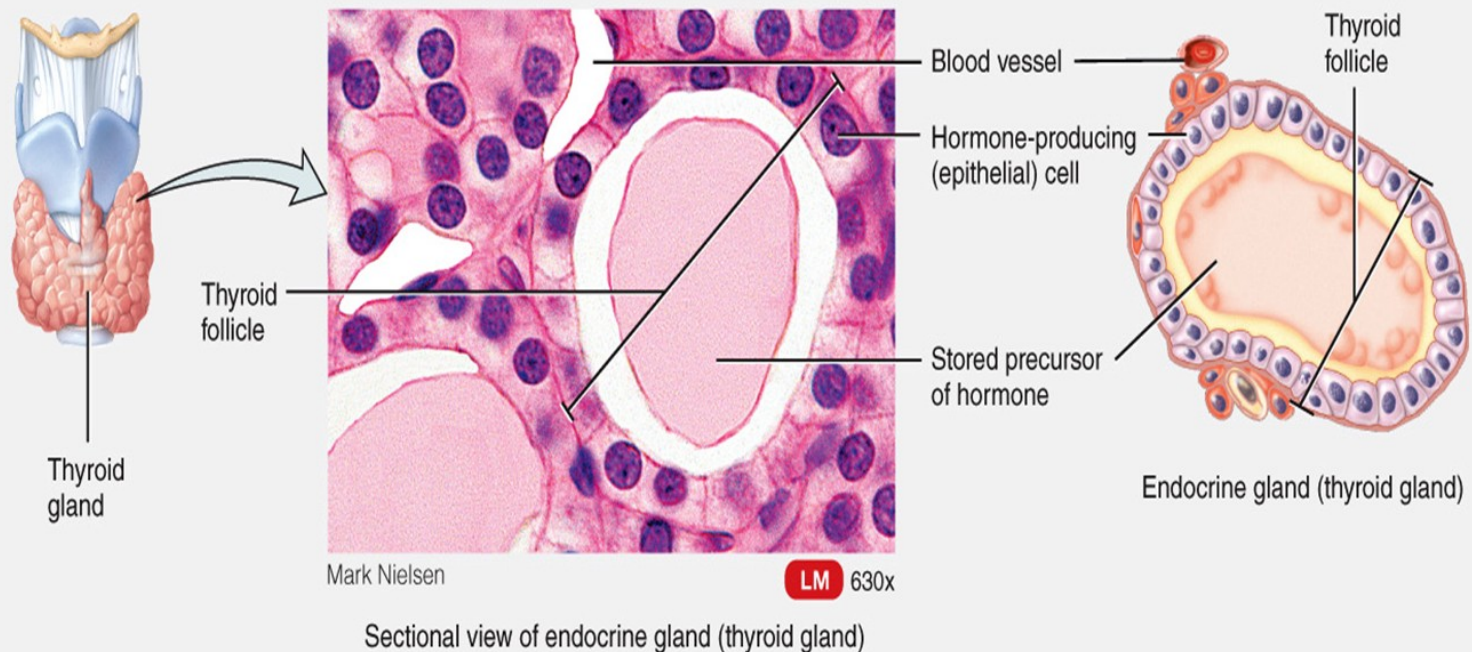
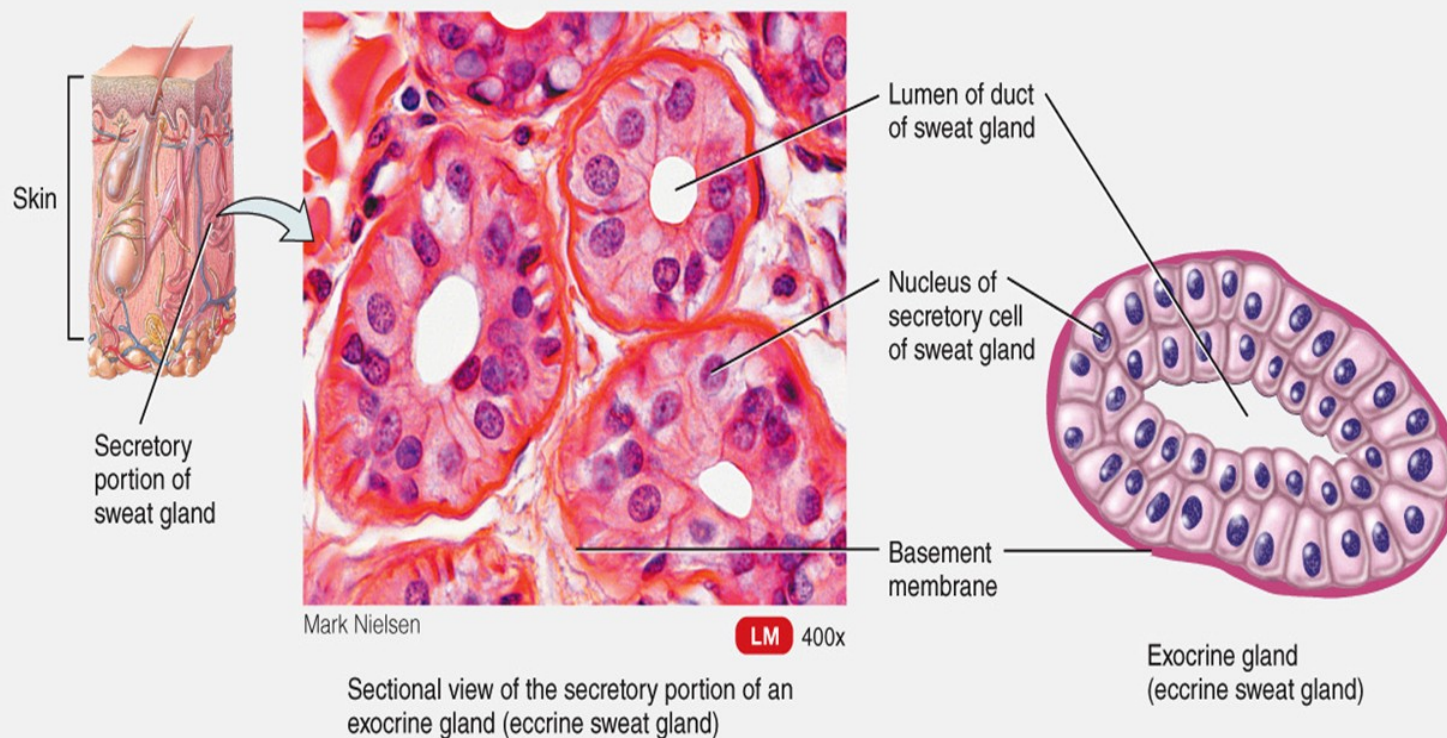


TABLE 4.2 Epithelial Tissue: Glandular Epithelium (Continued)

B. EXOCRINE GLANDS

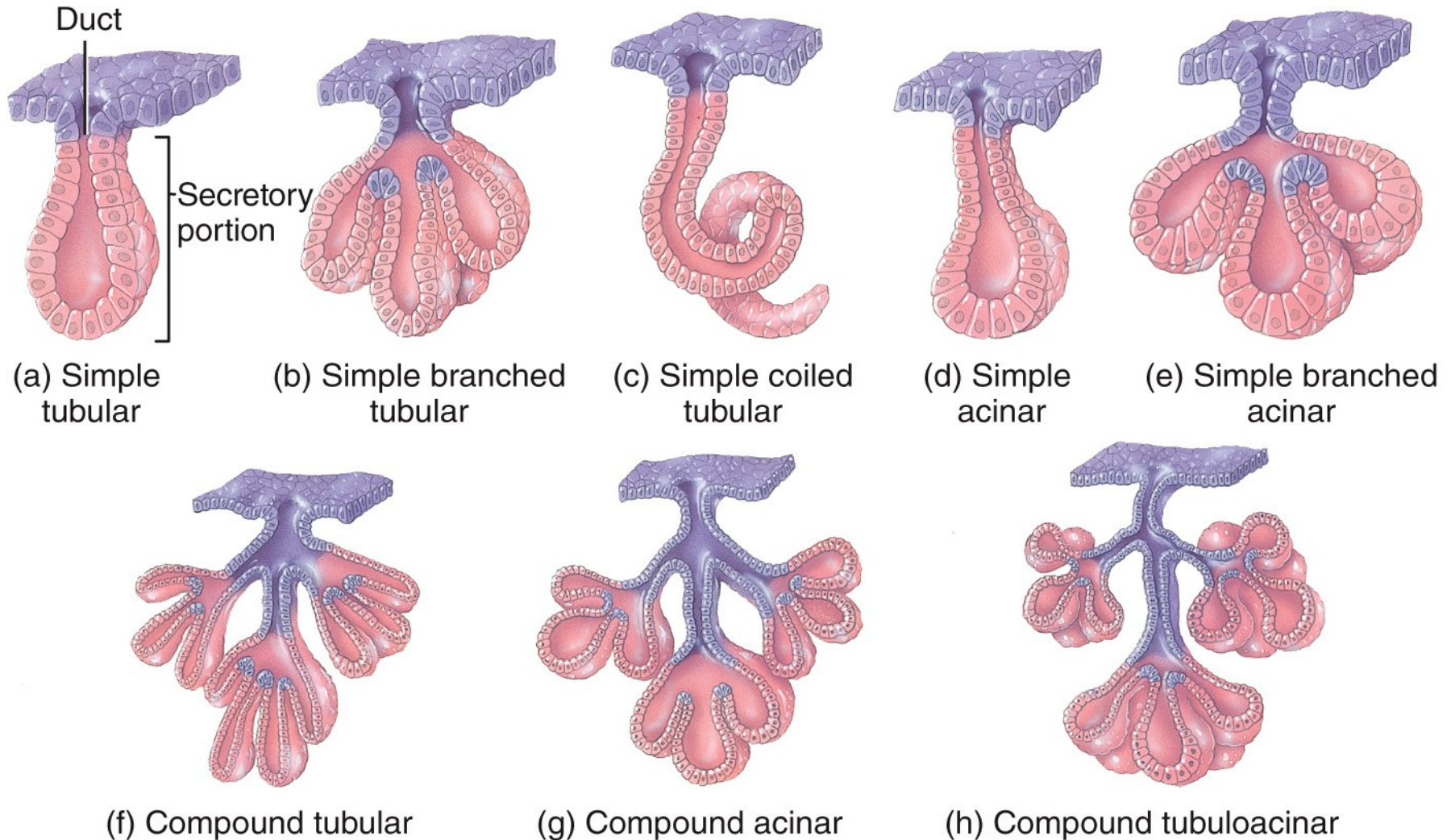
Description	Exocrine gland secretory products are released into ducts that empty onto surface of a covering and lining epithelium, such as skin surface or lumen of hollow organ.
Location	Sweat, oil, and earwax glands of skin; digestive glands such as salivary glands (secrete into mouth cavity) and pancreas (secretes into small intestine).
Function	Produce substances such as sweat to help lower body temperature, oil, earwax, saliva, or digestive enzymes.



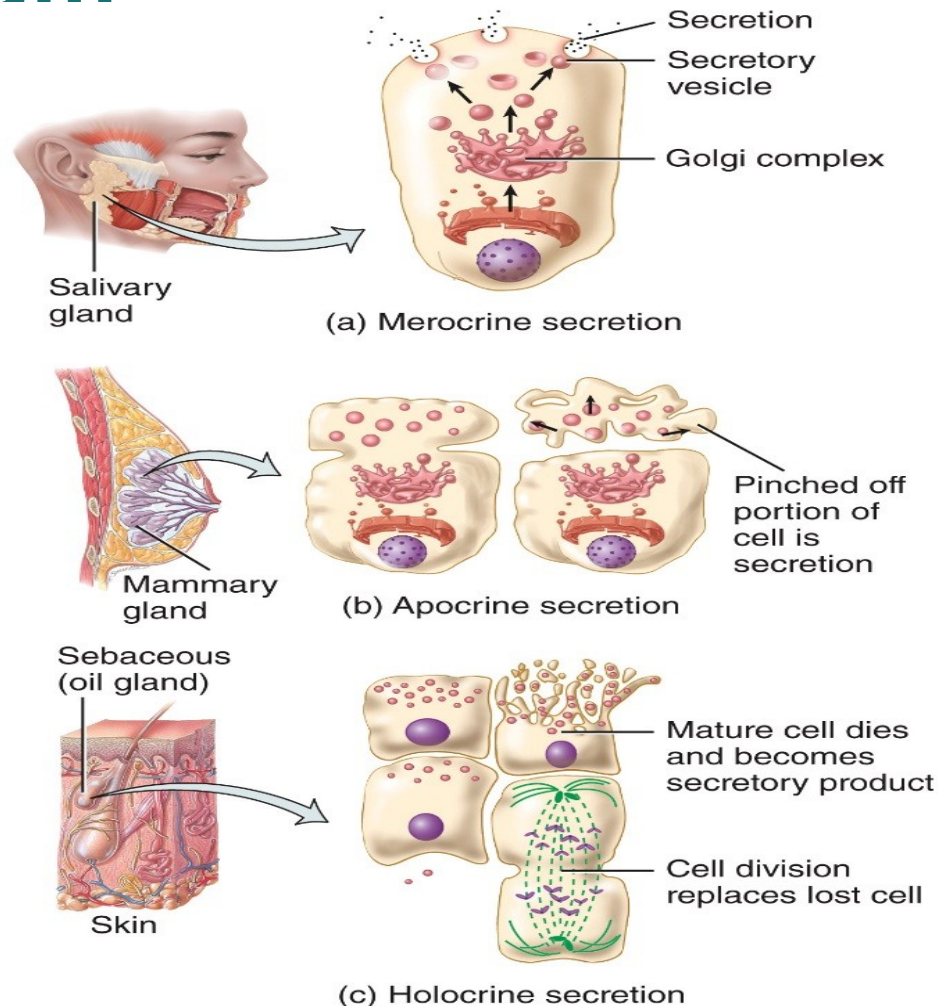
Structural Classification of Glandular Epithelium

- Unicellular – single cells
 - Goblet cells
- Multicellular – composed of many cells that form a distinctive microscopic structure or macroscopic organ
 - Sweat glands
 - Oil glands
 - Salivary glands

Multicellular Exocrine Glands



Functional Classification of Glandular Epithelium



Connective Tissue

General Features of Connective Tissue

- Consists of two basic elements:
 - Cells
 - Extracellular matrix
- Connective tissue cells do not have any free surfaces
- Connective tissue is highly vascularized and has a nerve supply
 - Exceptions: tendons and cartilage

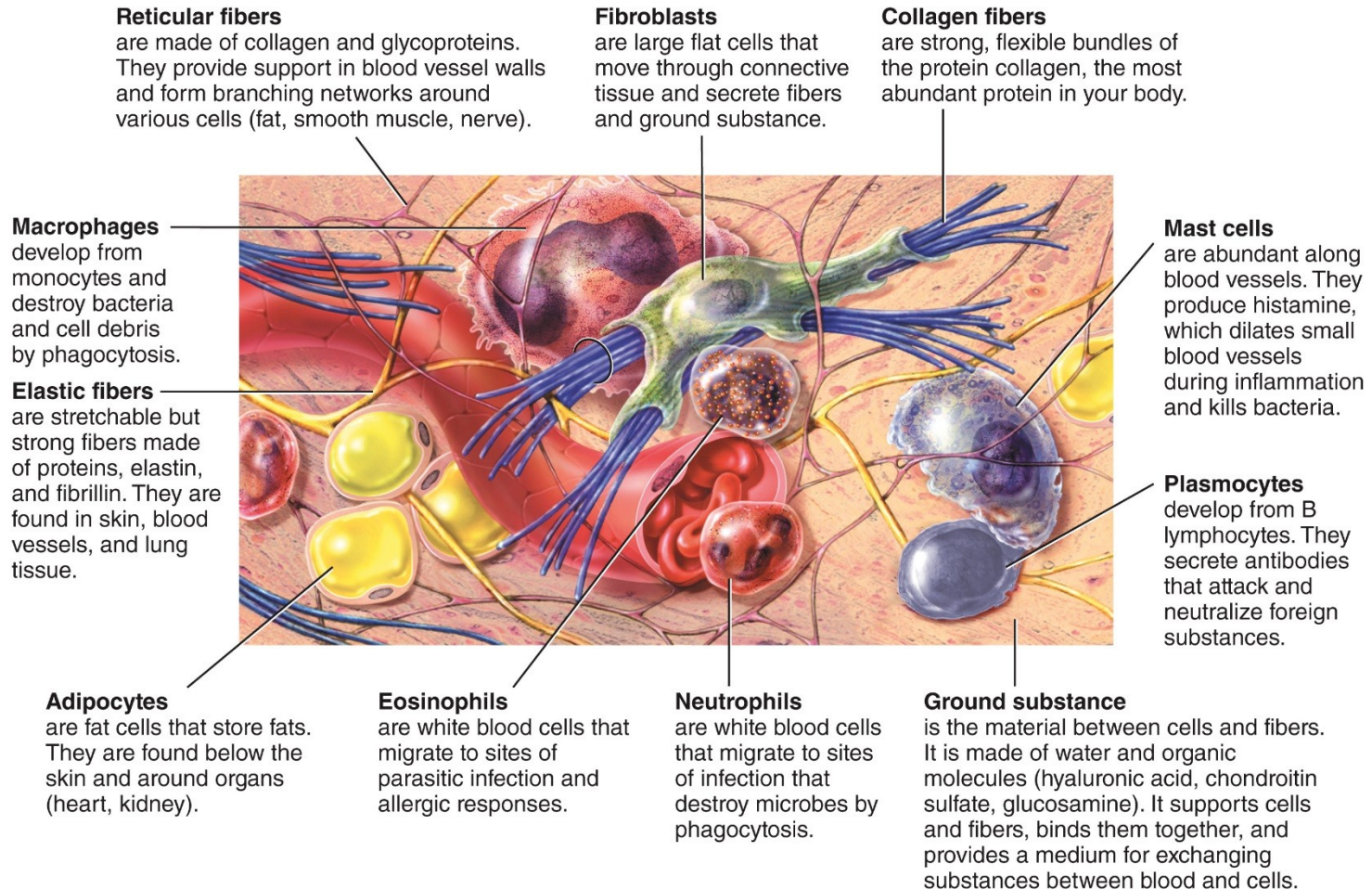
Connective Tissue Cells

- Fat cells, white blood cells, mast cells, and macrophages
- Mitotically active and secretory cells
 - “-blasts” = secretory cells; “-cytes” = mature cells
 - Fibroblasts in connect tissue proper
 - Chondroblasts and chondrocytes in cartilage
 - Osteoblasts and osteocytes in bone

Extracellular Matrix

- **Fibers**
 - Collagen fibers – strongest and most abundant type
 - Elastic fibers – thin, elastin fibers that allow for stretch
 - Reticular fibers – short, fine, branched collagenous fibers
- **Ground substance**
 - Fluid portions with various other components, depending on type
 - Can range from very watery (blood) to extremely solid (bone)

Connective Tissue Cells



Classification of Connective Tissue

TABLE 4.3 Embryonic Connective Tissues

A. MESENCHYME

Description	Mesenchyme has irregularly shaped mesenchymal cells embedded in semifluid ground substance that contains delicate reticular fibers.
Location	Almost exclusively under skin and along developing bones of embryo; some in adult connective tissue, especially along blood vessels.
Function	Forms almost all other types of connective tissue.

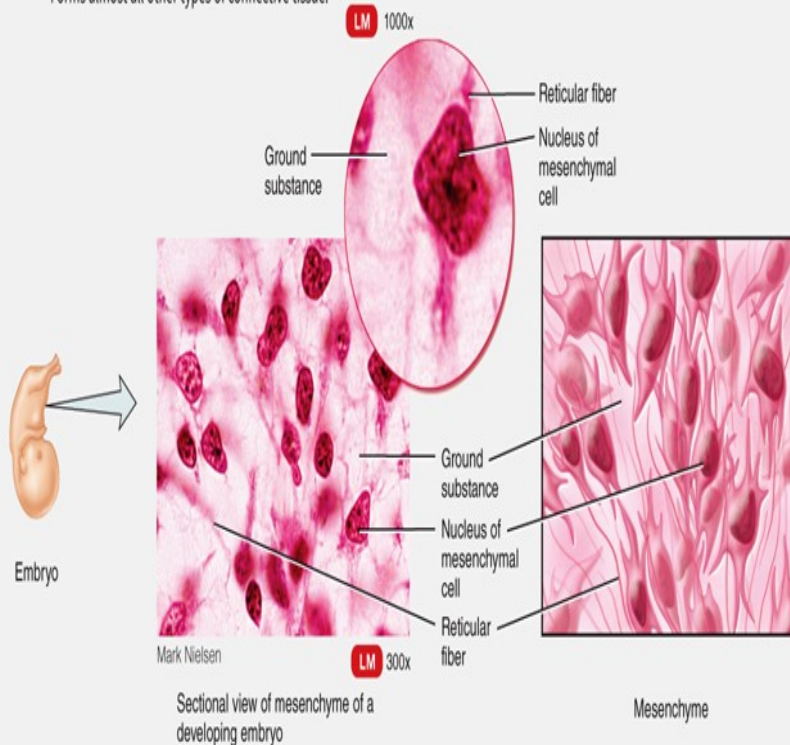
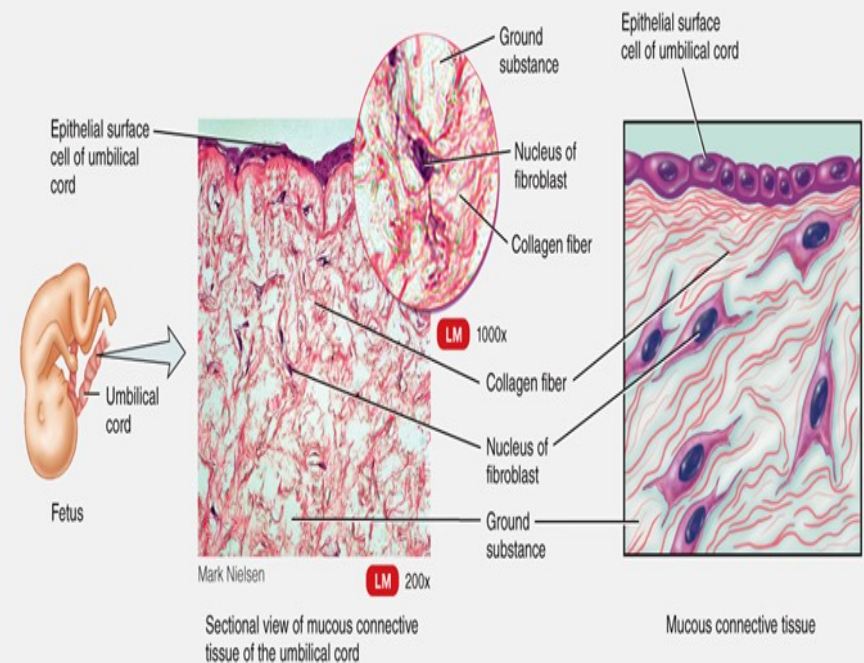


TABLE 4.3 Embryonic Connective Tissues

B. MUCOUS (MUOID) CONNECTIVE TISSUE

Description	Mucous (muoid) connective tissue has widely scattered fibroblasts embedded in viscous, jellylike ground substance that contains fine collagen fibers.
Location	Umbilical cord of fetus.
Function	Support.



Classification of Connective Tissue

- Mature
 - Loose
 - Dense
 - Cartilage
 - Bone
 - Blood

Connective Tissue Anatomy Overview:

Connective Tissues

TABLE 4.4 Mature Connective Tissue: Connective Tissue Proper—Loose Connective Tissue

A. AREOLAR CONNECTIVE TISSUE

Description	Areolar connective tissue is one of the most widely distributed connective tissues; consists of fibers (collagen, elastic, reticular) arranged randomly and several kinds of cells (fibroblasts, macrophages, plasma cells, adipocytes, mast cells, and a few white blood cells) embedded in semifluid ground substance (hyaluronic acid, chondroitin sulfate, dermatan sulfate, and keratan sulfate).
Location	In and around nearly every body structure (thus, called “packing material” of the body): in subcutaneous layer deep to skin; papillary (superficial) region of dermis of skin; lamina propria of mucous membranes; around blood vessels, nerves, and body organs.
Function	Strength, elasticity, support.

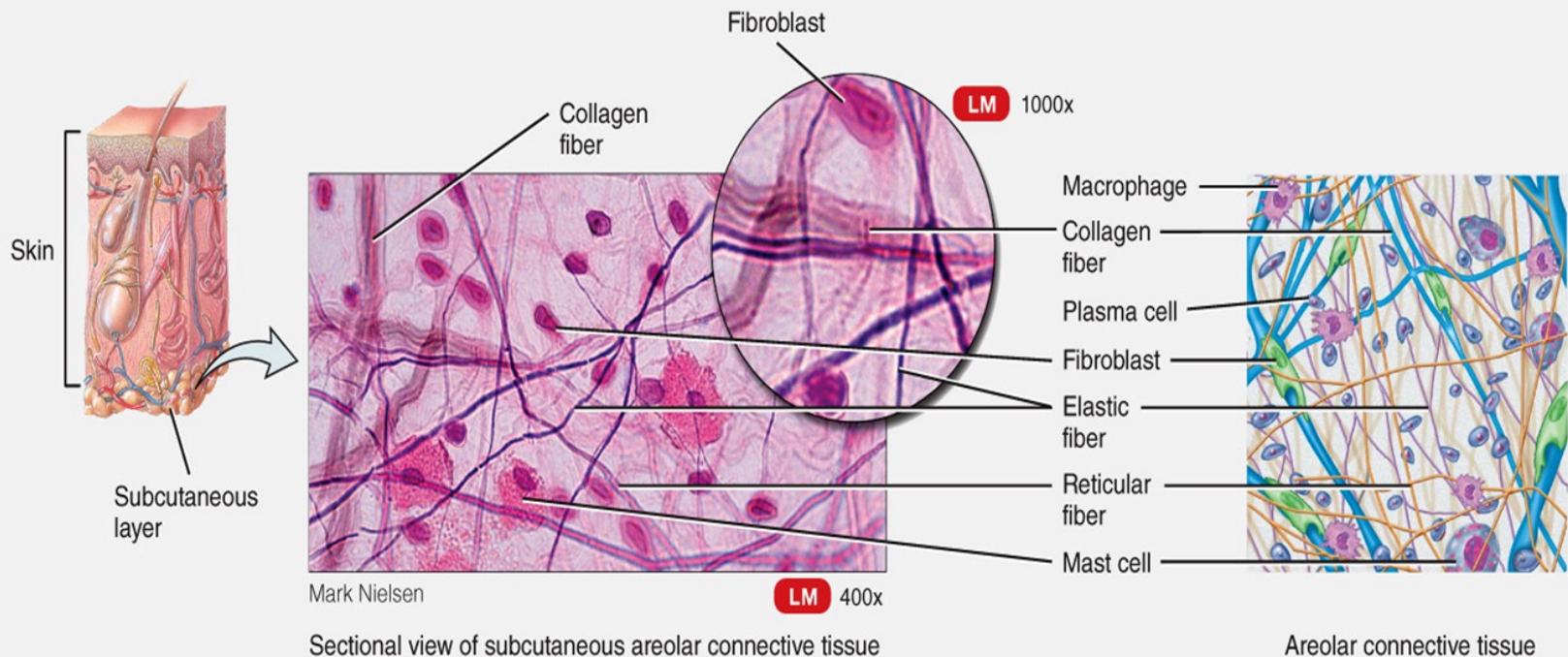
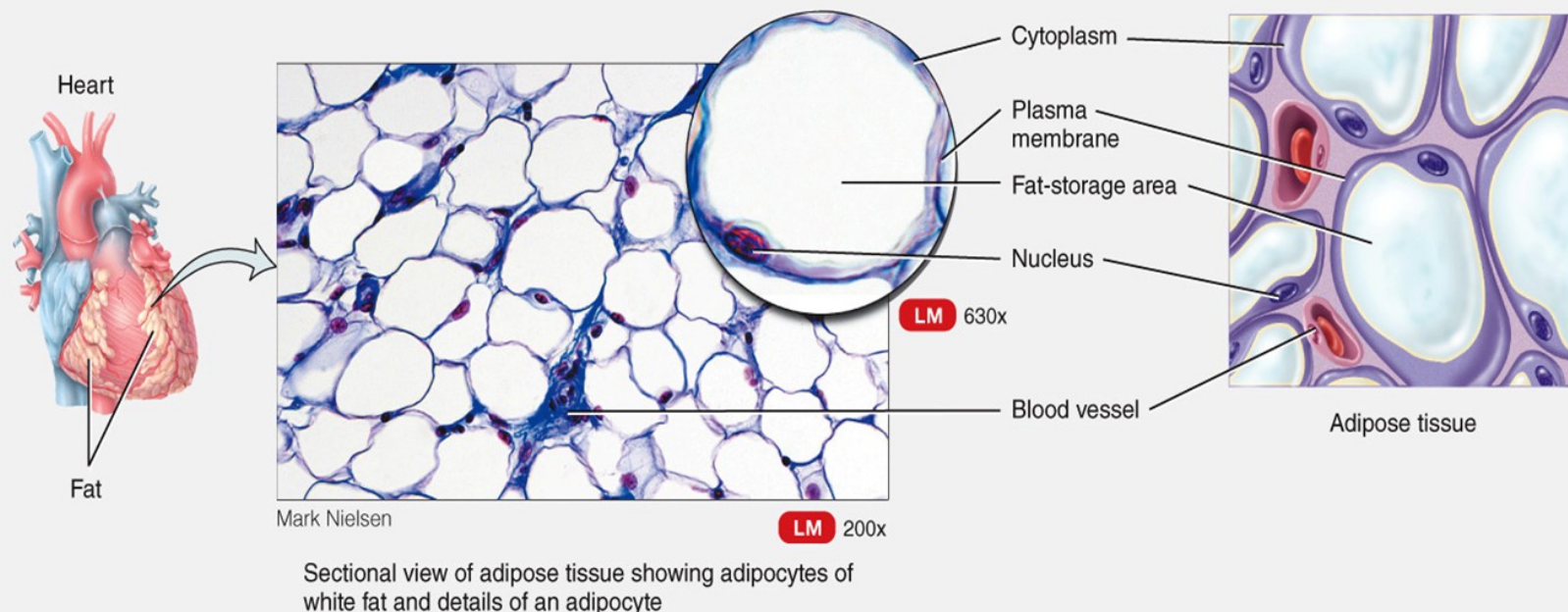


TABLE 4.4 Mature Connective Tissue: Connective Tissue Proper—Loose Connective Tissue

B. ADIPOSE TISSUE

Description	Adipose tissue has cells derived from fibroblasts (called <i>adipocytes</i>) that are specialized for storage of triglycerides (fats) as a large, centrally located droplet. Cell fills up with a single, large triglyceride droplet, and cytoplasm and nucleus are pushed to periphery of cell. With weight gain, amount of adipose tissue increases and new blood vessels form. Thus, an obese person has many more blood vessels than does a lean person, a situation that can cause high blood pressure, since the heart has to work harder. Most adipose tissue in adults is <i>white adipose tissue</i> (just described). <i>Brown adipose tissue</i> (BAT) is darker due to very rich blood supply and numerous pigmented mitochondria that participate in aerobic cellular respiration. BAT is widespread in the fetus and infant; adults have only small amounts.
Location	Wherever areolar connective tissue is located: subcutaneous layer deep to skin, around heart and kidneys, yellow bone marrow, padding around joints and behind eyeball in eye socket.
Function	Reduces heat loss through skin; serves as an energy reserve; supports and protects organs. In newborns, BAT generates heat to maintain proper body temperature. Adipose tissue is also an excellent source of stem cells, which are used in rejuvenation medicine to repair or replace damaged tissue.



C. RETICULAR CONNECTIVE TISSUE

Description	Reticular connective tissue is a fine interlacing network of reticular fibers (thin form of collagen fiber) and reticular cells.
Location	Stroma (supporting framework) of liver, spleen, lymph nodes; red bone marrow; reticular lamina of basement membrane; around blood vessels and muscles.
Function	Forms stroma of organs; binds smooth muscle tissue cells; filters and removes worn-out blood cells in spleen and microbes in lymph nodes.

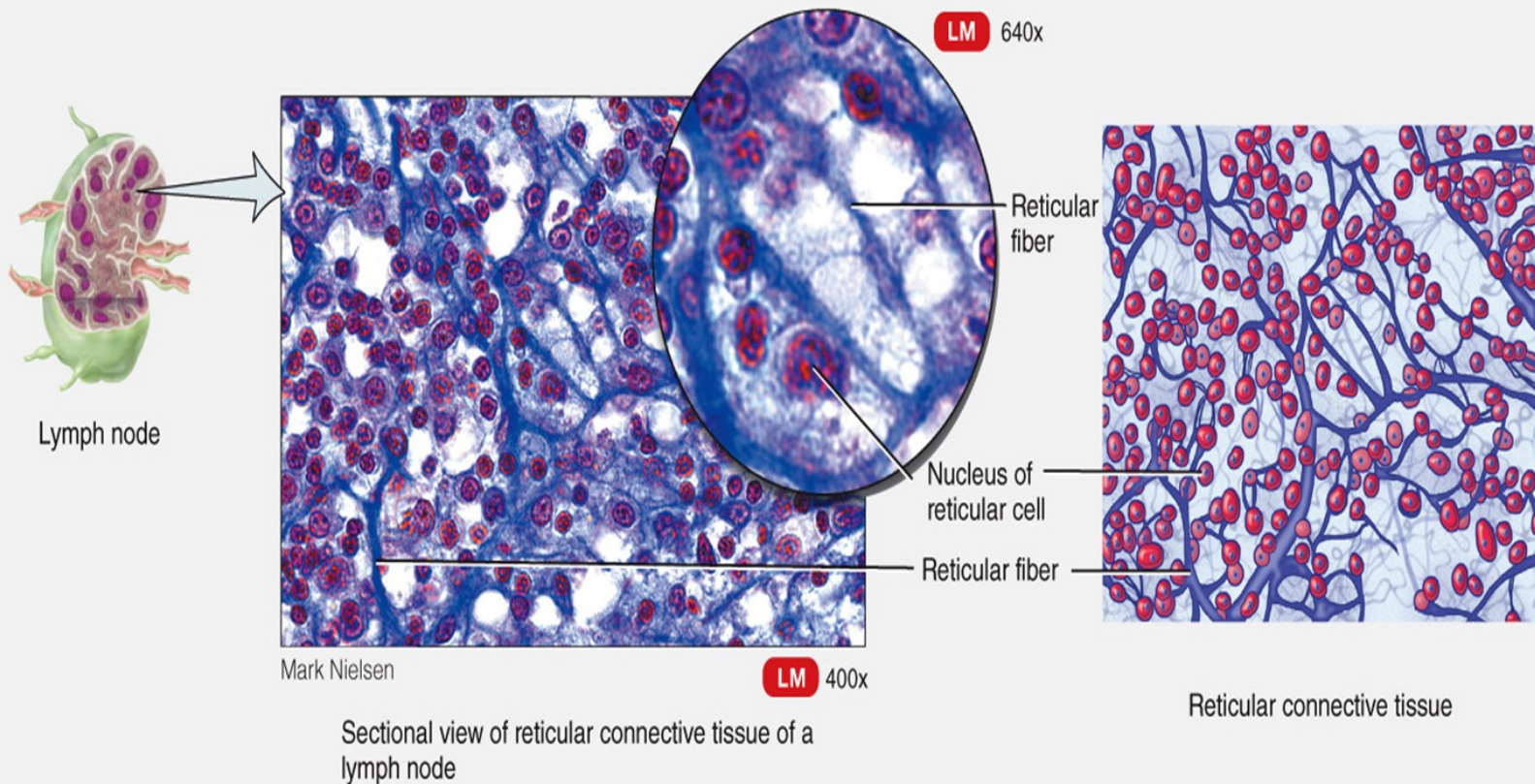


TABLE 4.5 Mature Connective Tissue: Connective Tissue Proper—Dense Connective Tissue

A. DENSE REGULAR CONNECTIVE TISSUE

Description	Dense regular connective tissue forms shiny white extracellular matrix; mainly collagen fibers <i>regularly</i> arranged in bundles with fibroblasts in rows between them. Collagen fibers (protein structures secreted by fibroblasts) are not living, so damaged tendons and ligaments heal slowly.
Location	Forms tendons (attach muscle to bone), most ligaments (attach bone to bone), and aponeuroses (sheetlike tendons that attach muscle to muscle or muscle to bone).
Function	Provides strong attachment between various structures. Tissue structure withstands pulling (tension) along long axis of fibers.

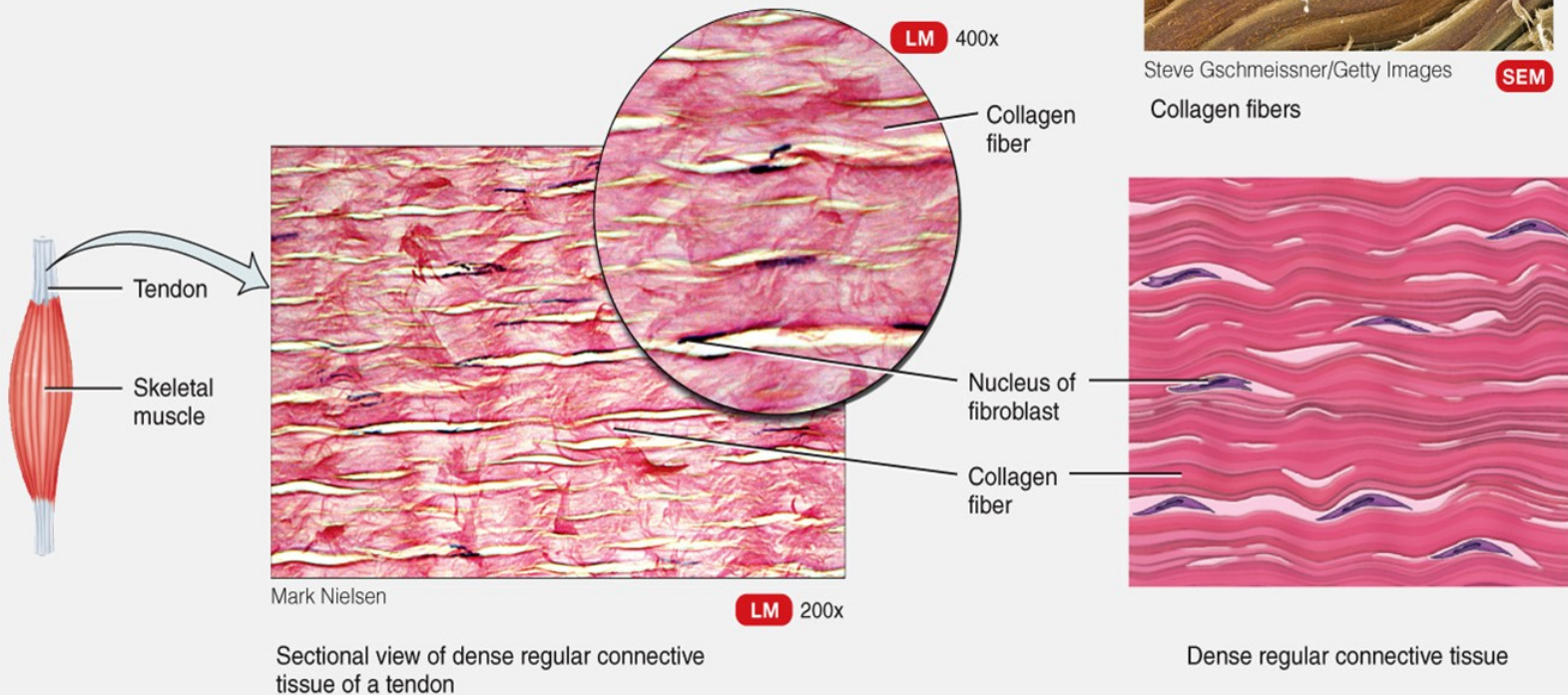
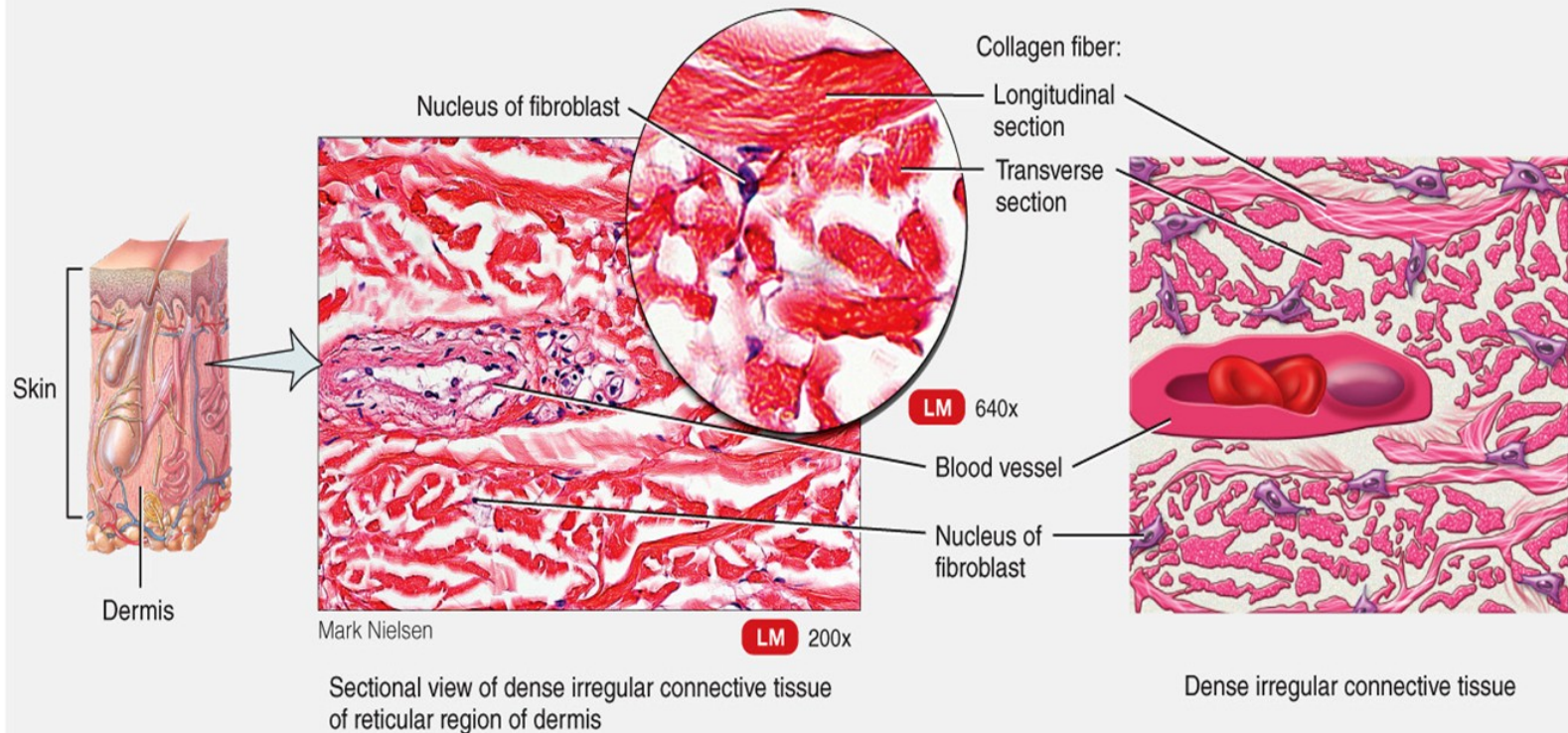


TABLE 4.5 Mature Connective Tissue: Connective Tissue Proper—Dense Connective Tissue

B. DENSE IRREGULAR CONNECTIVE TISSUE

Description	Dense irregular connective tissue is made up of collagen fibers; usually <i>irregularly</i> arranged with a few fibroblasts.
Location	Often occurs in sheets, such as fasciae (tissue beneath skin and around muscles and other organs), reticular (deeper) region of dermis of skin, fibrous pericardium of heart, periosteum of bone, perichondrium of cartilage, joint capsules, membrane capsules around various organs (kidneys, liver, testes, lymph nodes); also in heart valves.
Function	Provides tensile (pulling) strength in many directions.



C. ELASTIC CONNECTIVE TISSUE

Description	Elastic connective tissue contains predominantly elastic fibers with fibroblasts between them; unstained tissue is yellowish.
Location	Lung tissue, walls of elastic arteries, trachea, bronchial tubes, true vocal cords, suspensory ligaments of penis, some ligaments between vertebrae.
Function	Allows stretching of various organs; is strong and can recoil to original shape after being stretched. Elasticity is important to normal functioning of lung tissue (recoils in exhaling) and elastic arteries (recoil between heartbeats to help maintain blood flow).

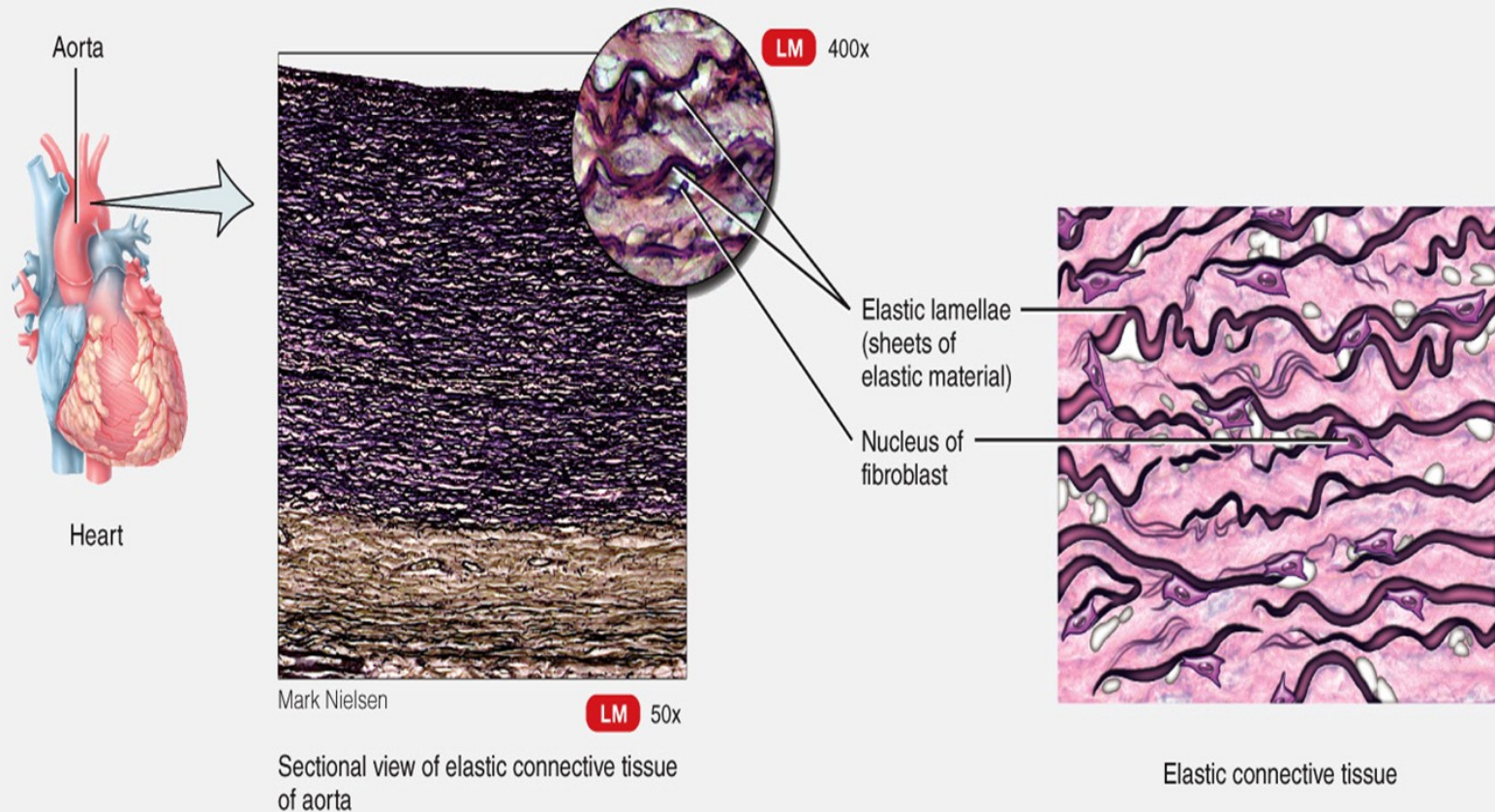
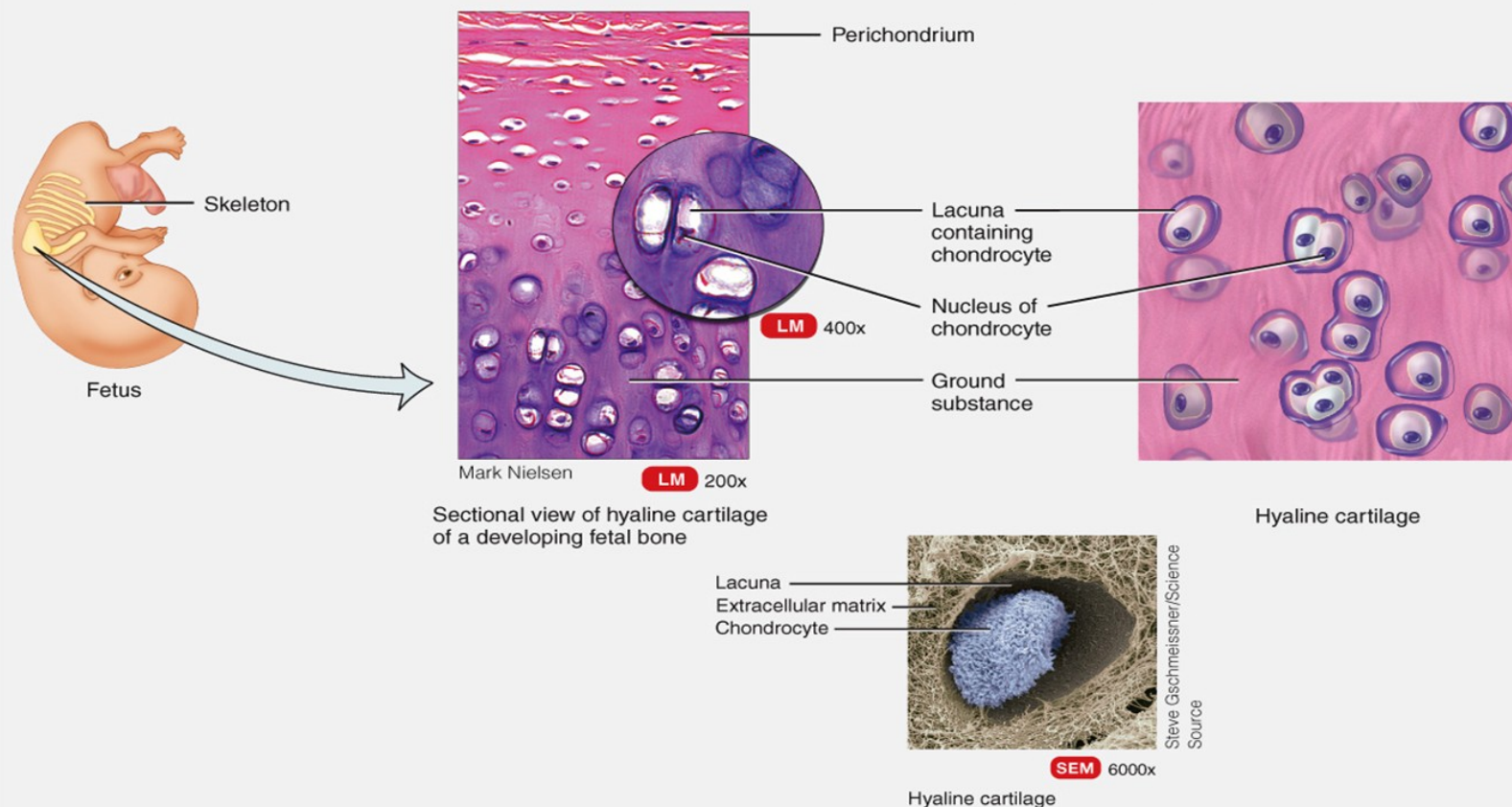


TABLE 4.6 Mature Connective Tissue: Supporting Connective Tissue—Cartilage

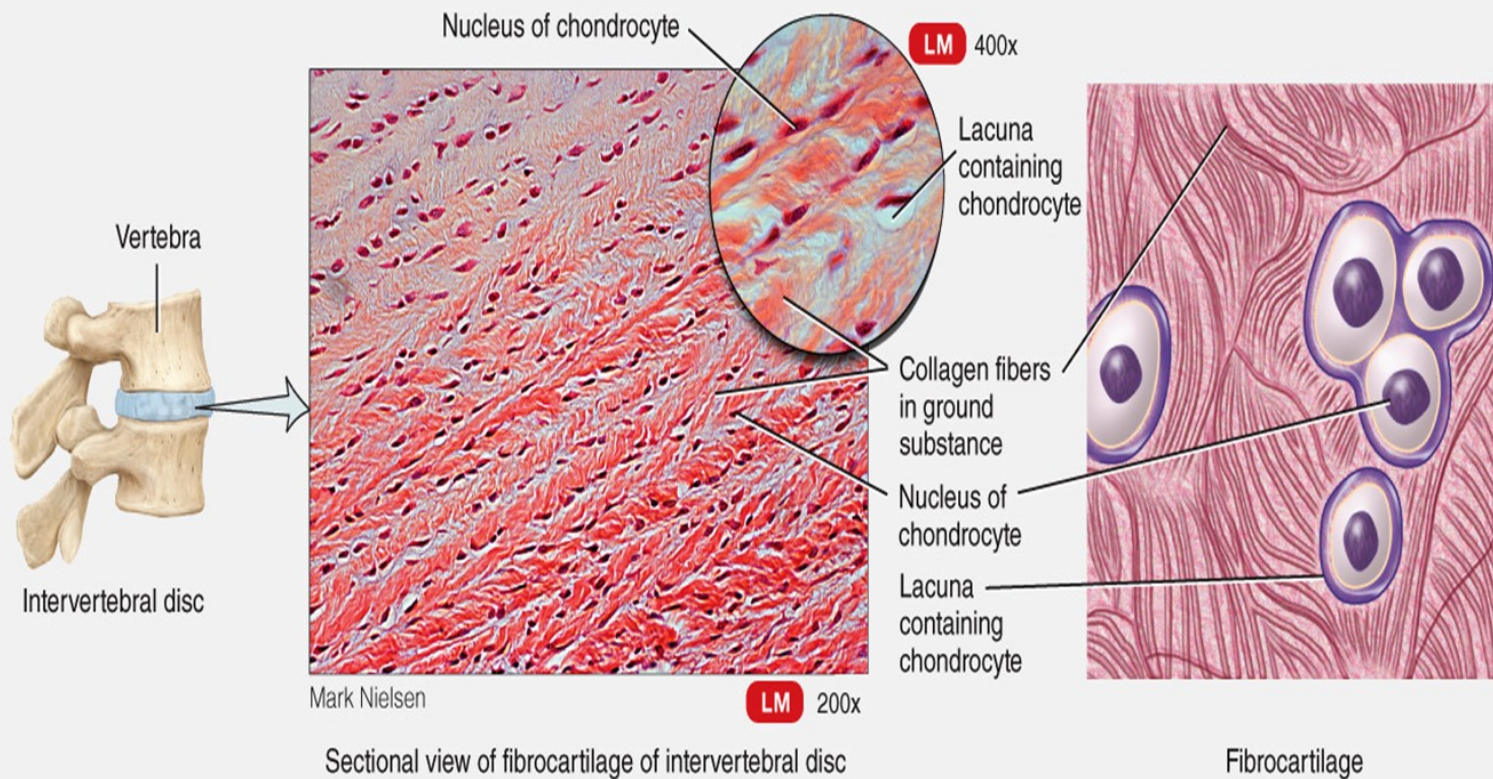
A. HYALINE CARTILAGE

Description	Hyaline cartilage (<i>hyalinus</i> = glassy) contains a resilient gel as ground substance and appears in the body as a bluish-white, shiny substance (can stain pink or purple when prepared for microscopic examination; fine collagen fibers are not visible with ordinary staining techniques); prominent chondrocytes are found in lacunae surrounded by perichondrium (exceptions: articular cartilage in joints and cartilage of epiphyseal plates, where bones lengthen during growth).
Location	Most abundant cartilage in body; at ends of long bones, anterior ends of ribs, nose, parts of larynx, trachea, bronchi, bronchial tubes, embryonic and fetal skeleton.
Function	Provides smooth surfaces for movement at joints, flexibility, and support; weakest type of cartilage and can be fractured.



B. FIBROCARTILAGE

Description	Fibrocartilage has chondrocytes among clearly visible thick bundles of collagen fibers within extracellular matrix; lacks perichondrium.
Location	Pubic symphysis (where hip bones join anteriorly), intervertebral discs, menisci (cartilage pads) of knee, portions of tendons that insert into cartilage.
Function	Support and joining structures together. Strength and rigidity make it the strongest type of cartilage.



C. ELASTIC CARTILAGE

Description	Elastic cartilage has chondrocytes in threadlike network of elastic fibers within extracellular matrix; perichondrium present.
Location	Lid on top of larynx (epiglottis), part of external ear (auricle), auditory (eustachian) tubes.
Function	Provides strength and elasticity; maintains shape of certain structures.

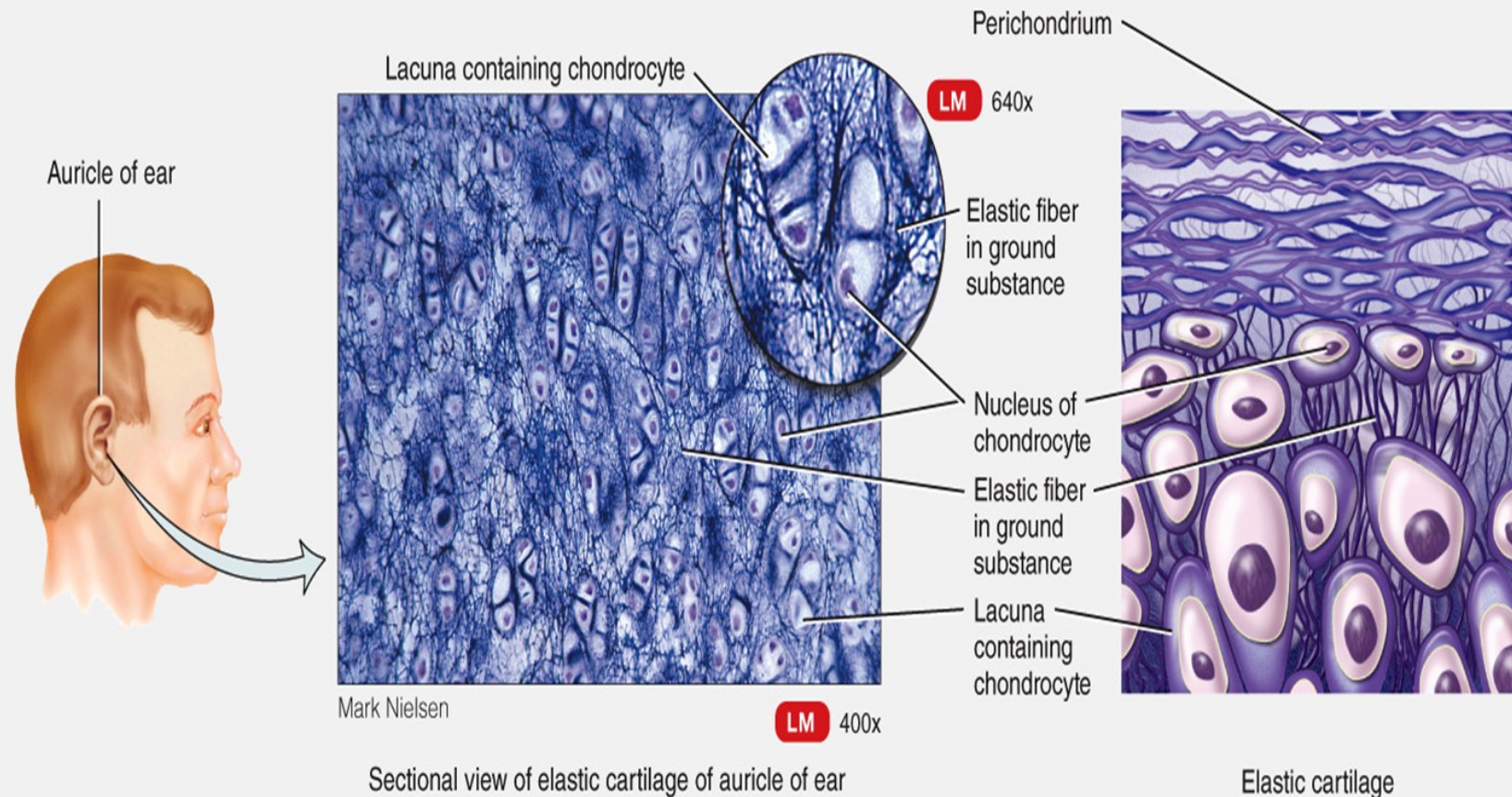


TABLE 4.8 Mature Connective Tissue: Liquid Connective Tissue—Blood

Description	Blood consists of blood plasma and formed elements: red blood cells (erythrocytes), white blood cells (leukocytes), platelets (thrombocytes).
Location	Within blood vessels (arteries, arterioles, capillaries, venules, veins), within chambers of heart.
Function	Red blood cells: transport oxygen and some carbon dioxide; white blood cells: carry on phagocytosis and mediate allergic reactions and immune system responses; platelets: essential for blood clotting.

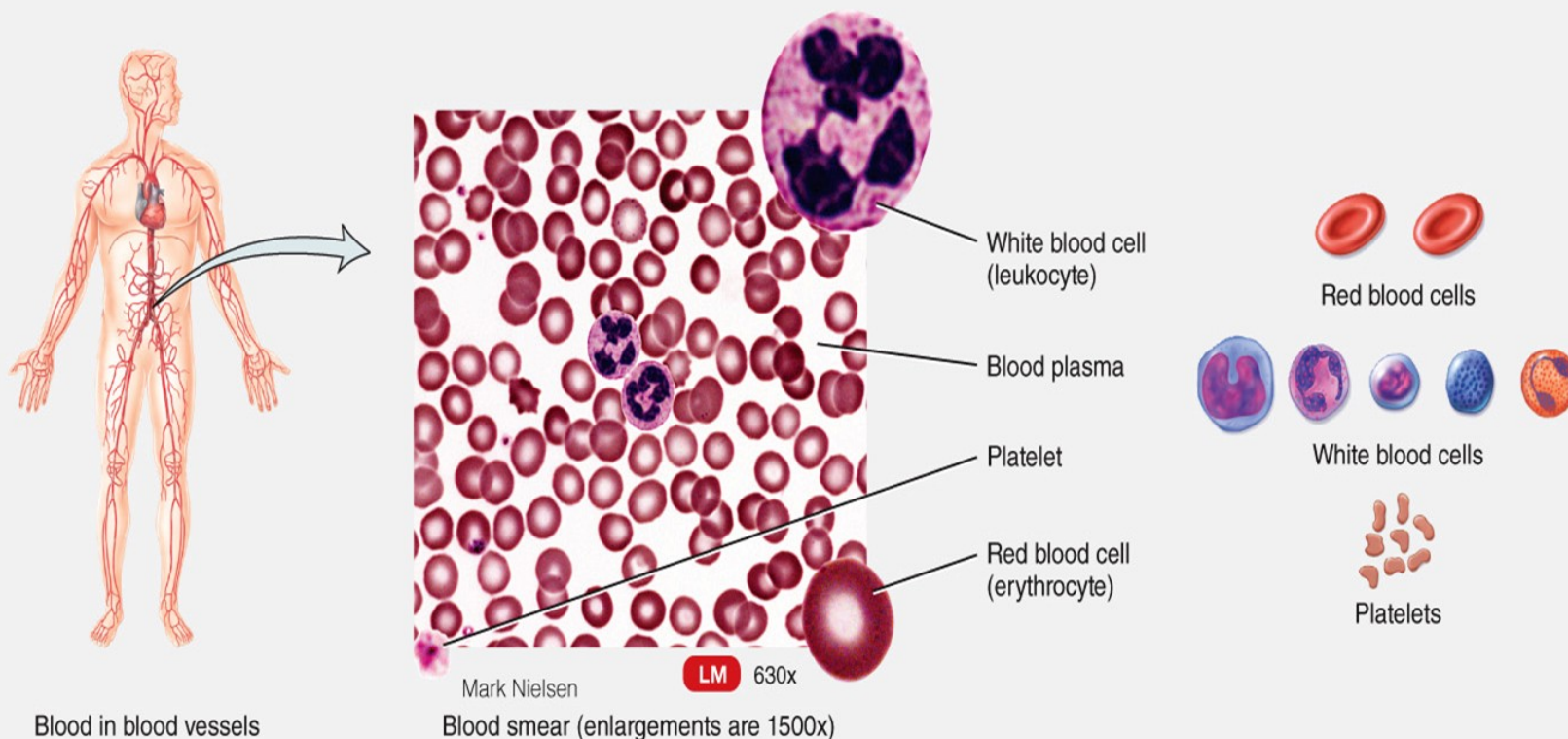
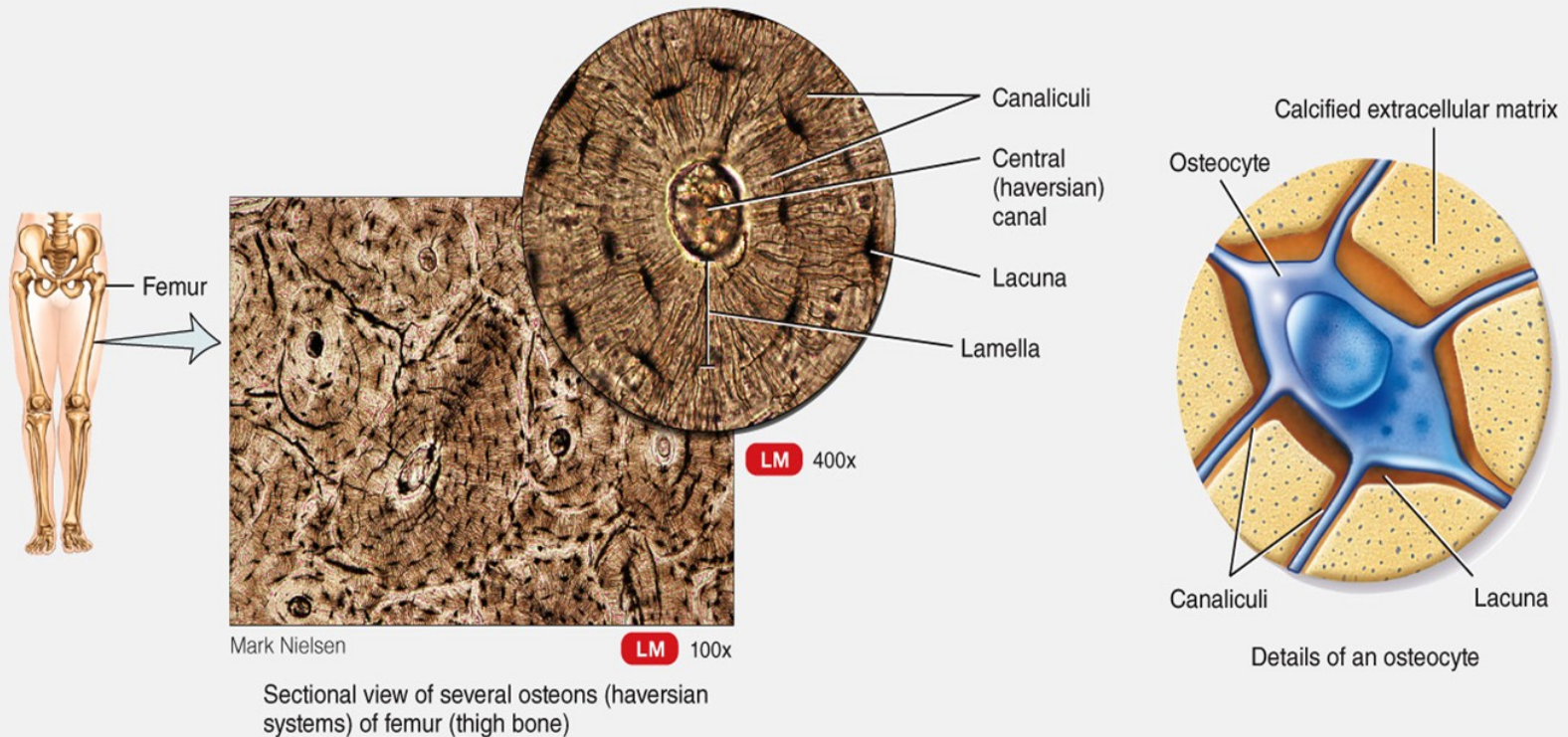


TABLE 4.7 Mature Connective Tissue: Supporting Connective Tissue—Bone Tissue

Description	Compact bone tissue consists of osteons (haversian systems) that contain lamellae, lacunae, osteocytes, canaliculi, and central (haversian) canals. By contrast, spongy bone tissue (see Figure 6.3) consists of thin columns called trabeculae; spaces between trabeculae are filled with red bone marrow.
Location	Both compact and spongy bone tissue make up the various parts of bones of the body.
Function	Support, protection, storage; houses blood-forming tissue; serves as levers that act with muscle tissue to enable movement.

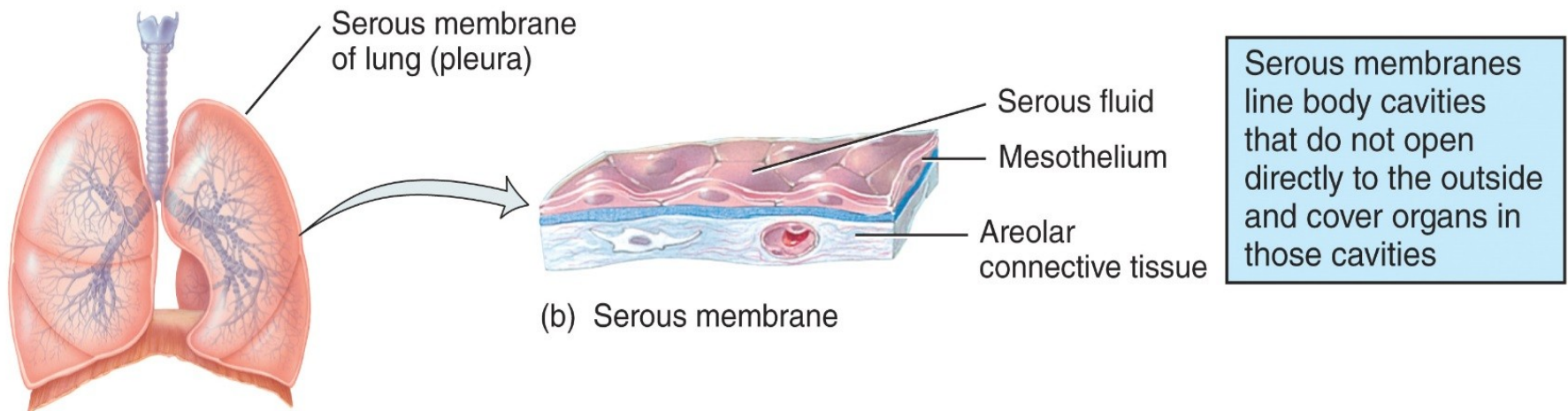
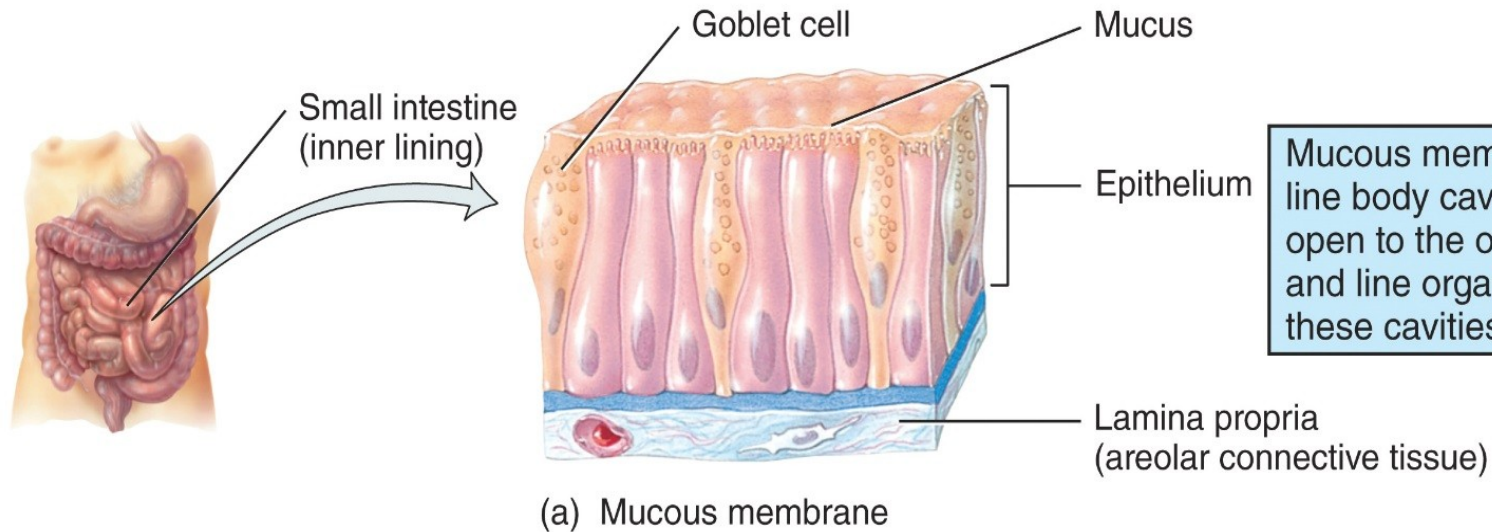


Membranes

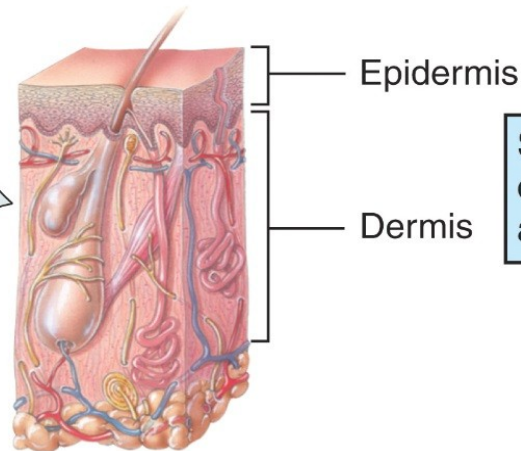
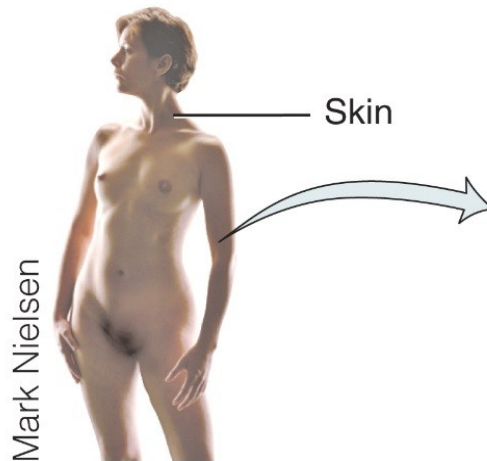
Membranes

- Membranes are flat sheets of pliable tissue that cover or line a part of the body
- 2 types of membranes
 - Epithelial membranes
 - Mucous membranes
 - Serous membranes
 - Cutaneous membranes
 - Synovial membranes

Membranes



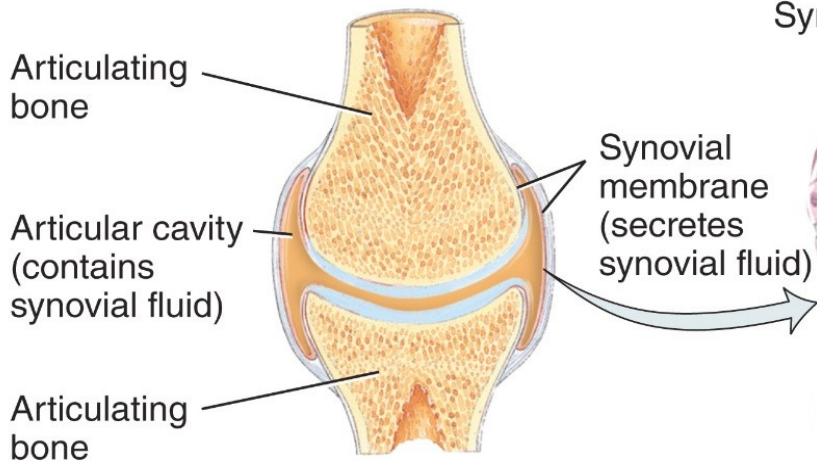
Membranes



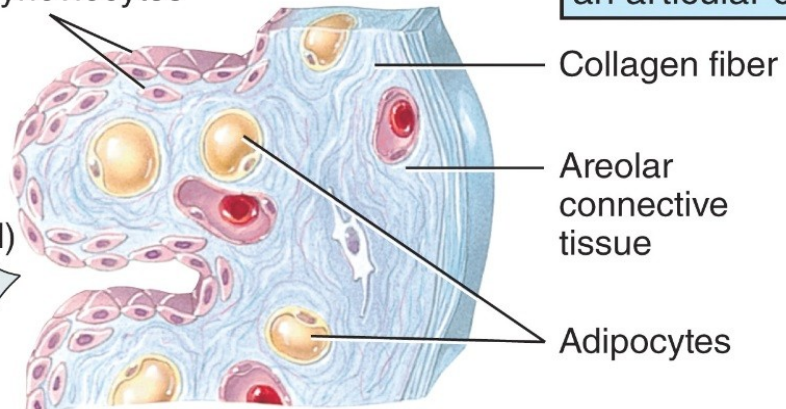
Skin covers the surface of the body that contain an articular cavity

(c) Skin (cutaneous membrane)

Synovial membranes line joints that contain an articular cavity.



Synoviocytes



(d) Synovial membrane

Muscular Tissue

General Characteristics of Muscular Tissue

- Muscular tissue consists of fibers that provide motion, maintain posture, and produce heat
- Well vascularized
- Muscle cells are also called muscle fibers
- 3 types of muscle tissue:
 - Skeletal muscle
 - Cardiac muscle
 - Smooth muscle

Muscle Tissue

Anatomy Overview:

Muscular Tissue

TABLE 4.9 Muscular Tissue

A. SKELETAL MUSCLE TISSUE

Description	Skeletal muscle tissue consists of long, cylindrical, striated fibers (<i>striations</i> are alternating light and dark bands within fibers that are visible under a light microscope). Skeletal muscle fibers vary greatly in length, from a few centimeters in short muscles to 30–40 cm (about 12–16 in.) in the longest muscles. A muscle fiber is a roughly cylindrical, multinucleated cell with nuclei at the periphery. Skeletal muscle is considered <i>voluntary</i> because it can be made to contract or relax by conscious control.
Location	Usually attached to bones by tendons.
Function	Motion, posture, heat production, protection.

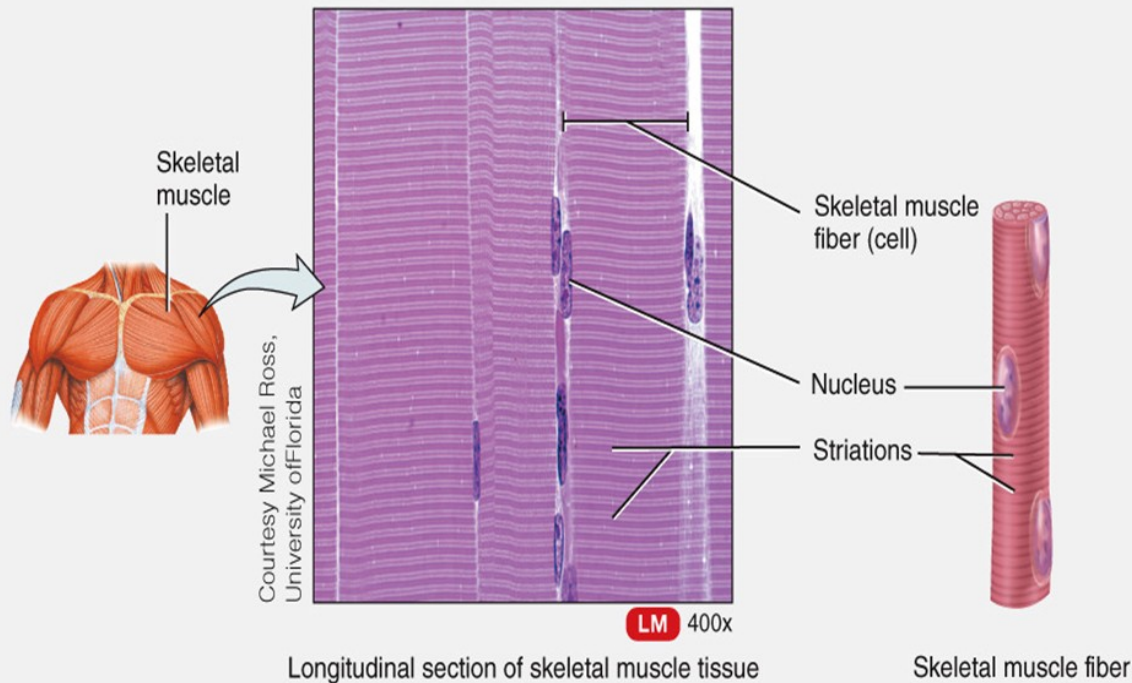
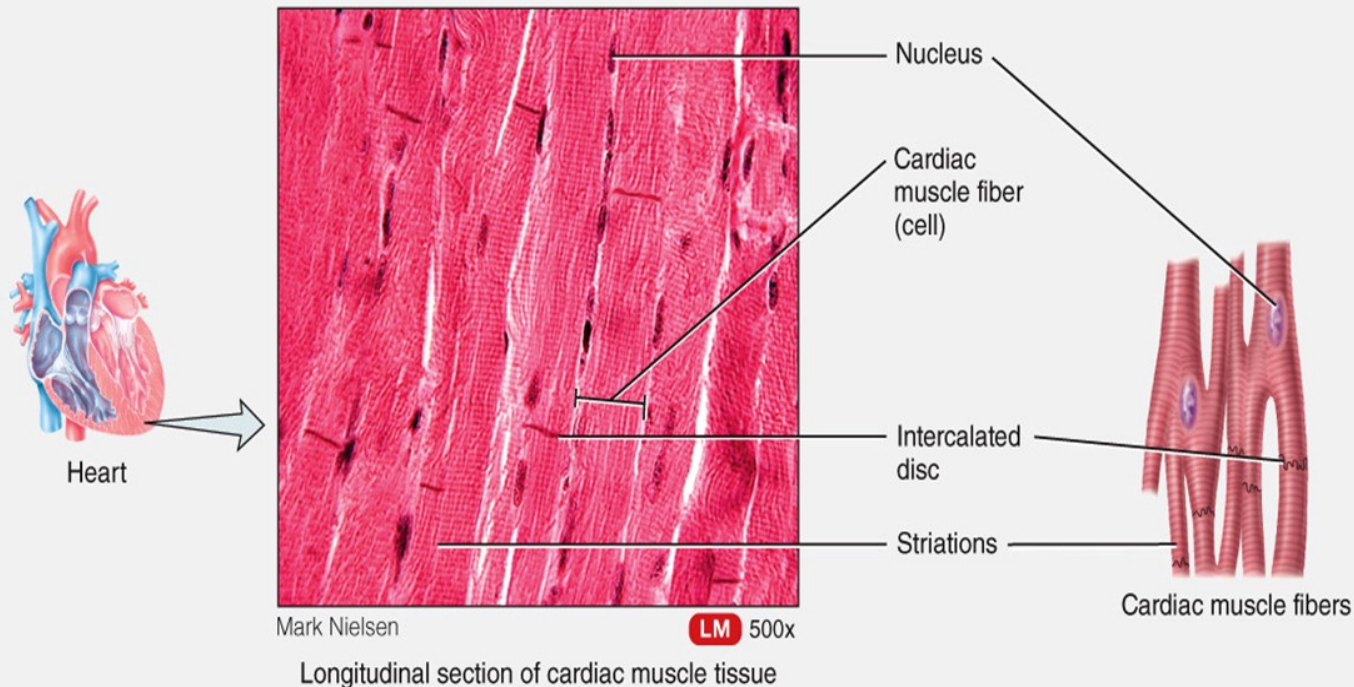


TABLE 4.9 Muscular Tissue

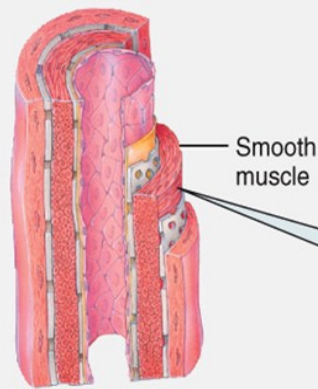
B. CARDIAC MUSCLE TISSUE

Description	Cardiac muscle tissue consists of branched, striated fibers with usually only one centrally located nucleus (occasionally two). Attach end to end by transverse thickenings of plasma membrane called <i>intercalated discs</i> (in-TER-ka-lāt-ed; <i>intercalate</i> = to insert between), which contain desmosomes and gap junctions. Desmosomes strengthen tissue and hold fibers together during vigorous contractions. Gap junctions provide route for quick conduction of electrical signals (muscle action potentials) throughout heart. <i>Involuntary</i> (not conscious) control.
Location	Heart wall.
Function	Pumps blood to all parts of body.

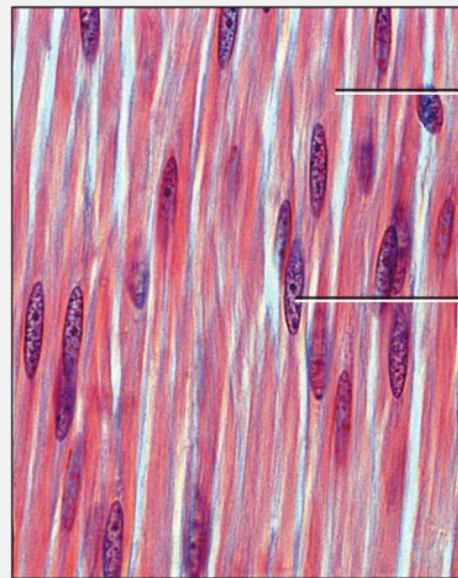


C. SMOOTH MUSCLE TISSUE

Description	Smooth muscle tissue consists of nonstriated fibers (lack striations, hence the term <i>smooth</i>). Smooth muscle fiber is a small spindle-shaped cell thickest in middle, tapering at each end, and containing a single, centrally located nucleus. Gap junctions connect many individual fibers in some smooth muscle tissue (for example, in wall of intestines). Usually involuntary; can produce powerful contractions as many muscle fibers contract in unison. Where gap junctions are absent, such as iris of eye, smooth muscle fibers contract individually, like skeletal muscle fibers.
Location	Iris of eyes; walls of hollow internal structures such as blood vessels, airways to lungs, stomach, intestines, gallbladder, urinary bladder, and uterus.
Function	Motion (constriction of blood vessels and airways, propulsion of foods through gastrointestinal tract, contraction of urinary bladder and gallbladder).



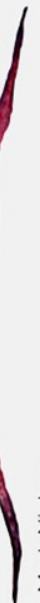
Artery



Mark Nielsen

LM 500x

Longitudinal section of smooth muscle tissue



Smooth muscle fiber

Nervous Tissue

General Characteristics of Nervous Tissue

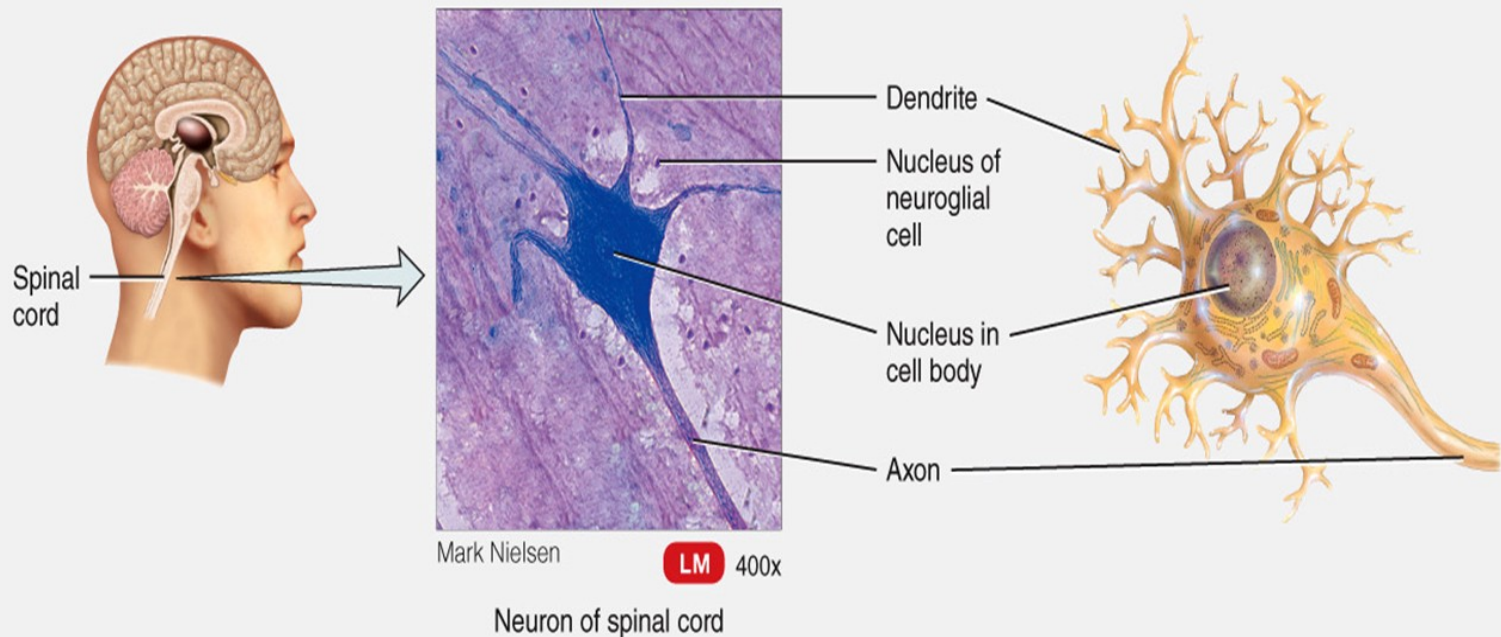
- Two kinds of cells:
 - Neurons
 - Have cell body, dendrites, and axons
 - Can carry sensory or motor information and they can perform integrative functions
 - Generate and transmit nervous impulses
 - Neuroglia
 - Protect and support neurons
 - Do not generate or transmit nervous impulses
 - Also called glial cells

Nervous Tissue Anatomy Overview:

Nervous Tissue

TABLE 4.10 Nervous Tissue

Description	Nervous tissue consists of (1) neurons (nerve cells), which consist of cell body and processes extending from cell body (one to multiple dendrites and a single axon); and (2) neuroglia, which do not generate or conduct nerve impulses but have other important supporting functions.
Location	Nervous system.
Function	Exhibits sensitivity to various types of stimuli; converts stimuli into nerve impulses (action potentials); conducts nerve impulses to other neurons, muscle fibers, or glands.



Excitable Cells

Excitable Cells

- Neurons and muscle fibers are considered excitable cells because they exhibit electrical excitability
 - Electrical excitability is the ability to respond to certain stimuli by producing electrical signals, such as action potentials, which travel along the plasma membrane of a neuron or muscle fiber due to the presence of specific voltage-gated channels

Tissue Repair: Restoring Homeostasis

Tissue Repair

- Tissue repair is the process that replaces worn out, damaged, or dead cells
 - Epithelial cells are replaced by the division of stem cells or undifferentiated cells
 - Not all connective tissue cells have the ability to repair
 - Muscle cells can perform limited repair
 - Some nervous cells can perform limited repair, others cannot
- Fibrosis is the formation of scar tissue (does not function like original tissue)

Aging and Tissues

Aging

- Younger bodies generally experience:
 - A better nutritional state
 - A better blood supply to tissues
 - A faster metabolic rate
- Aging slows the process of tissue repair
- Aging also results in stiffening and loss of elasticity in tissues

Homeostatic Imbalances

- Disorders of epithelial tissues tend to be specific to individual organs, such as ulcers in the stomach
- Disorders of connective tissue examples
 - Some are autoimmune in nature, such as lupus
 - Osteogenesis imperfecta (abnormal bone formation) and Ehlers-Danlos syndrome (hyperflexible joints and stretchy skin)
- Disorders of muscular and nervous tissues will be discussed in later chapters

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